Research Report

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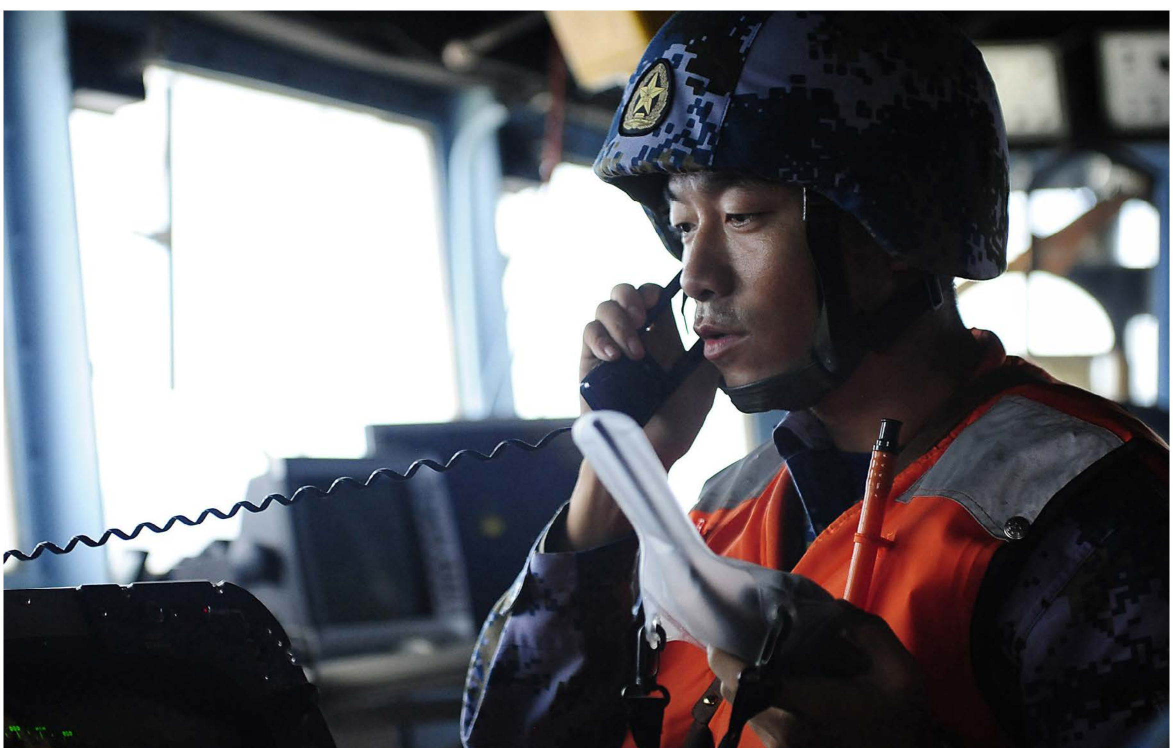
The People's Liberation Army Navy's Approach to Maintenance Management

Cooking Dumplings in a Teapot



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About This Report

The ability for navies to procure, field, and sustain large surface combatants is central to modern power projection. The U.S. Navy's approach to ship procurement coupled with infrastructure issues and workforce shortages at U.S. shipyards has led to significant delays in both naval shipbuilding and maintenance activities. Tensions between high operational demands and the need to conduct training and maintenance to sustain ship readiness levels have resulted in significant mishaps and crashes. The ability of the People's Liberation Army Navy (PLAN) to sustain its growing inventories of more-capable systems will rely on its logistics capabilities and processes. The PLAN's rapid modernization has created a larger burden for the logistics support and maintenance of PLAN assets. Although the PLAN has focused on improving its maintenance self-sufficiency, organic maintenance units still appear to depend on higher maintenance echelons to conduct more-sophisticated repairs. Overall, the PLAN will likely take a hybrid approach to maintenance going forward via efforts to improve its maintenance organization, processes, and culture to repair and sustain its most-capable assets while replacing some older classes of ships.

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Summary

The ability for navies to procure, field, and sustain large surface combatants is central to modern power projection. The U.S. Navy's approach to ship procurement coupled with infrastructure issues and workforce shortages at U.S. shipyards has led to significant delays in both naval shipbuilding and maintenance activities. Tensions between high operational demands and the need to conduct training and maintenance to sustain ship readiness levels have resulted in significant mishaps and crashes. The ability of the People's Liberation Army Navy (PLAN) to sustain its growing inventories of more-capable systems will rely on its logistics capabilities and processes. Understanding the PLAN's approach to maintenance management is a necessary step in assessing the People's Liberation Army's (PLA's) ability to sustain a modernized force, and there are implications for both competition and conflict.

Research Questions

- How does the rapid growth and modernization of the PLAN inform maintenance management and activities?
- What approaches do other navies take to balance maintenance demand and supply, and how does the PLAN's approach compare?
- How is maintenance controlled and organized to meet maritime-specific requirements?
- What factors shape the PLAN's ability to maintain and sustain its forces?

Approach

We first sought to understand the interrelationship between shipbuilding and naval maintenance. We examined three case studies—the modern U.S. and Japanese navies, along with the Cold War Soviet Navy—to identify themes that could apply to the PLAN. To examine the details of PLAN maintenance activities, we reviewed PLA sources, primarily drawing from articles on PLAN maintenance topics featured in *PLA Daily*, and other PLA and Chinese state media and select non-authoritative Chinese media sources. In analyzing these maintenance activities, we leveraged a broader array of PLA sources—professional military education texts, journal articles, and technical and reference books on PLA naval equipment and naval maintenance term— and English-language secondary sources on related PLAN topics, such as training and organizational culture.

Key Findings

- The PLAN has moved from iteratively designing and producing small ship classes to series production of larger ship classes. Series production could help streamline maintenance requirements but has stressed the maintenance workforce and associated organizations, such as vessel training centers, which ensure ship deployability.
- If overhaul rates of newer ship classes remain similar to those for older destroyer classes, then overhauls of China's most-capable surface combatants will begin in 2031 and last through at least 2042.

- After comparing the PLAN with U.S., Japanese, and Soviet naval forces, we concluded that the PLAN is likely to take a hybrid approach as it continues to replace old force structure, invest in the skill of its maintenance forces, and pursue a repair-focused investment strategy.
- The PLAN has focused on improving its maintenance self-sufficiency, but organic maintenance units still appear to undertake surface-level maintenance practices and depend on higher maintenance echelons to conduct more-sophisticated repairs. This tendency could be further exacerbated if the PLAN continues to grow its force structure.
- Although the PLAN has promoted innovative problem-solving and sought to destigmatize mistakes, sources still emphasize dogmatic adherence to standards.
- Theater-specific demands, such as high operational tempos in the Eastern Theater Command Navy (TCN), could shape each TCN's maintenance management and activities.
- Morale issues might have an impact on retention, particularly for naval units that are shore-based rather than ship-based. Even experienced maintainers appear to work through exhaustion and illness to ensure that naval maintenance activities are completed.

Implications

Some of our findings reflect issues that other navies are similarly grappling with, such as the following:

- **Workforce issues.** As ships become more technologically sophisticated, the bar for maintainer proficiency continues to be raised.
- **Increasing costs.** The increasing cost and sophistication of technologies being designed into new ship classes and added to existing ones has increased per-ship cost and will result in a construction slowdown for the PLAN.
- **The changing nature of repair.** Practices are evolving from repairing components to replacing them with advanced components.

We assess four other areas as PLAN-specific dynamics:

- **The sheer volume of new ships complicates maintenance pipelines and cycles.** The net gain to fielded combatants has required the PLAN to increase maintainer efficiency and likely improve other support. These increased costs stack on top of the added expense of more-advanced components on new classes of surface vessels.
- **Some PLAN assets undergo significant changes, requiring maintainers to reskill or develop new resource materials from scratch.** Even the most-capable hero maintainers discussed needing to remap pipeline pathways after significant repairs. This additional burden might tie back to discussion of a repair model in which each system developer fulfills the specified requirements, but there is a lack of integration across stakeholders.
- **The PLAN's focus on achieving self-sufficiency could reduce overall efficiency.** Whether and how the PLAN decides to balance this trade-off remains to be seen.
- **Emphasis on repairing rather than replacing small components could prioritize reuse over efficiency.** If widespread, this emphasis could imply that time is spent on low-level repairs as opposed to such activities as more-sophisticated training opportunities or mentoring junior personnel.

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CHAPTER 1

Introduction

The ability of the People's Liberation Army (PLA) to sustain its growing inventories of more-capable systems will rely on its logistics capabilities, systems, and processes. Previous research has focused on the PLA Army (PLAA) and PLA Air Force (PLAAF),[1] but the PLA Navy (PLAN) faces its own unique maintenance challenges because naval forces require both underway maintenance and maintenance ashore. Understanding the PLAN's approach to maintenance management is a necessary step in assessing the PLA's ability to sustain a modernized force, and there are implications for both competition and conflict.

What Is at Stake

The ability for navies to procure, field, and sustain large surface combatants is central to modern power projection.[2] The U.S. Navy's (USN's) approach to ship procurement coupled with infrastructure issues and workforce shortages at U.S. shipyards has led to significant delays in both naval shipbuilding and maintenance activities.[3] Tensions between high operational demands and the need to conduct training and maintenance to sustain USN ship readiness levels have resulted in significant incidents, such as four ships from Seventh Fleet experiencing mishaps and crashes in 2017.[4] More broadly, rising costs for operations and sustainment across the U.S. armed forces further constrain the Department of the Navy from fielding and maintaining large ships employed for a variety of missions during competition and conflict.[5] Given that the USN is seeking to modernize its capabilities while continuing to field many aging platforms,[6] some analysts have posited that the PLAN might have a comparative advantage in fielding advanced warships, aided by China's

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1 Joslyn Fleming, Cristina L. Garafola, Elisa Yoshiara, Sale Lilly, and Alexis Dale-Huang, *Kicking the Tires? The People's Liberation Army's Approach to Maintenance Management*, RAND Corporation, RR-A1995-1, 2023.

2 Jonathan Masters, *Sea Power: The U.S. Navy and Foreign Policy*, Council on Foreign Relations, June 12, 2024.

3 U.S. Government Accountability Office (GAO), *Shipbuilding and Repair: Navy Needs a Strategic Approach for Private Sector Industrial Base Investments*, GAO-25-106286, February 2025a; GAO, *Navy Shipbuilding: Enduring Challenges Call for Systemic Change*, GAO-25-108225, March 2025b.

4 Megan Eckstein, "Five Years Later: Inside the Navy's Data-Driven Quest to Avert a Future Fitzgerald or McCain Collision," *Navy Times*, June 16, 2022.

5 Doug Berenson, "Getting the Defense Department off the Hamster Wheel: Reducing Operating Costs to Invest in the Future of the Force," *War on the Rocks*, March 31, 2021. On USN life-cycle management processes and practices, see Bradley Martin, Roland J. Yardley, Phillip Pardue, Brynn Tannehill, Emma Westerman, and Jessica Duke, *An Approach to Life-Cycle Management of Shipboard Equipment*, RAND Corporation, RR-2510-NAVY, 2018.

6 See, for example, Caitlyn Burchett, "12 Aging Navy Destroyers Due to Retire Will Now Have Their Service Extended," *Stars and Stripes*, October 31, 2024; and Megan Eckstein, "Boxer Deployment Delay Highlights Aging Fleet, Lack of Repair Capacity," *Defense News*, March 2, 2024.

world-leading shipbuilding industry.[7] China also might benefit from lower sustainment costs of its younger naval fleet and from a greater capacity to repair ships.[8] However, these types of assessments do not take into account how the growing technological sophistication of PLAN ships could raise the bar for the types and quality of maintenance performed or whether the breakneck pace of ship production has had second- or third-order implications for maintenance management across the fleet.

Overall, little research has focused on the extent to which the PLA can sustain its forces. Some research has focused on PLA logistics distribution capabilities,[9] but, overall, the literature on PLA logistics—particularly on key subfunctions, such as maintenance—is lacking.[10] Exploration of PLAN maintenance has been especially limited.[11] Although some PLAN maintenance dynamics are similar to those of the PLAA and PLAAF, there are unique considerations related to the PLAN's ties to civilian industry, which shape its approach to military-civilian fusion. The PLAN also has more-substantive differences between maintenance that is conducted in port and maintenance that is conducted at sea, which can provide insights into power projection and sustainment capabilities. Recent reports reveal that technicians lack requisite skills that might limit the full use of naval platforms, indicating a gap in skill proficiency among the maintainer force that has been observed in other PLA services.[12]

Research Questions

Further exploration of maintenance issues affecting the PLAN could identify similarities with issues affecting the PLAA and PLAAF along with unique challenges for the PLAN:

- How is maintenance controlled and organized to meet maritime-specific requirements?
- How does the rapid growth and modernization of the PLAN inform maintenance management and activities?
- What approaches do other navies take to balance maintenance demand and supply, and how do the PLAN's approaches compare?
- What factors shape the PLAN's ability to maintain and sustain its forces?

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[7] Laura Bicker, "China Navy's Is Expanding at Breakneck Speed—and Catching Up with the US," BBC, August 31, 2025; Jerry Hendrix, "The Age of American Naval Dominance Is Over," *The Atlantic*, March 23, 2023.

[8] On capacity to repair during conflict, see Alexander Palmer, Henry H. Carroll, and Nicholas Velazquez, "Unpacking China's Naval Buildup," Center for Strategic & International Studies, June 5, 2024.

[9] Leigh Anne Luce and Erin Richter, "Handling Logistics in a Reformed PLA: The Long March Toward Joint Logistics," in Phillip C. Saunders, Arthur S. Ding, Andrew Scobell, Andrew N. D. Yang, and Joel Wuthnow, eds., *Chairman Xi Remakes the PLA: Assessing Chinese Military Reforms*, National Defense University Press, 2019; Joel Wuthnow, "A New Era for Chinese Military Logistics," *Asian Security*, Vol. 17, No. 3, February 2021; Conor M. Kennedy, "Logistics of the PLA Navy in Support of Operational Endurance," in Joshua Arostegui, ed., *The 2024 Carlisle Conference on the PLA: Protracted War Against the PRC*, U.S. Army War College Press, 2026.

[10] Fleming et al., 2023.

[11] Justin Boggess and Travis Dolney, "PLA Navy At-Sea Sustainment Capabilities," in Michael Shatzer and Roger Cliff, eds., *PLA Logistics and Sustainment: PLA Conference 2022*, U.S. Army War College Press, 2023. Related works include Kevin McCauley, *China Maritime Report No. 22: Logistics Support for a Cross-Strait Invasion: The View from Beijing*, China Maritime Studies Institute, 2022; and Ryan Martinson, *China Maritime Report No. 24: Incubators of Sea Power: Vessel Training Centers and the Modernization of the PLAN Surface Fleet*, China Maritime Studies Institute, November 2022.

[12] "Report: PLA Navy Runs into Crewing Difficulties for Growing Fleet," *Maritime Executive*, January 3, 2023; Kristin Huang, "Chinese Military Short of Troops Trained in Hi-Tech Operations, PLA Daily Reveals in Rare Show of Candour," *South China Morning Post*, January 2, 2023.

Research Approach and Scope

Exploratory Research

Similar to what we did in our 2023 report on the PLAA and PLAAF,[13] we performed foundational, exploratory research leveraging Chinese-language sources on PLAN maintenance topics. To guide our search, we leveraged insights from the 2023 report and from across RAND's recent and historical work on logistics and maintenance topics.[14] We also drew from Cold War research on the Soviet Navy and contemporary research on USN logistics and maintenance issues.[15]

Methodology

We first sought to understand the interrelationship between shipbuilding—a recent area of Chinese strength and U.S. struggles—and naval maintenance. We examined three case studies—the modern USN, the Cold War Soviet Navy, and the modern Japan Maritime Self-Defense Force (MSDF)—to identify themes that could apply to the PLAN. The appendix summarizes PLAN shipbuilding data for large surface combatants since 2000, which updates data provided in a 2024 RAND working paper on People's Republic of China (PRC) shipbuilding.[16]

To examine the details of PLAN maintenance activities, we reviewed PLA sources, primarily drawing from articles on PLAN maintenance topics featured in *PLA Daily*, and other PLA and Chinese state media and select non-authoritative Chinese media sources. We discuss these sources further in Chapter 3. In analyzing these maintenance activities, we leveraged a broader array of PLA sources—professional military education texts, journal articles, and technical and reference books on PLA naval equipment and naval maintenance terms—and English-language secondary sources on related PLAN topics, such as training and organizational culture.[17]

Scope

To focus our efforts, we limited the main scope of our analysis to PLAN surface combatant ships and supporting infrastructure. Although recent developments regarding submarine forces and personnel raise interesting questions about their maintenance and broader logistics support,[18] we lacked enough granular information to analyze submarine maintenance practices in detail. As of 2023, the PLAN's aviation force has

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13 Fleming et al., 2023.

14 RAND Corporation, "Military Logistics," webpage, undated.

15 On the Cold War, see James Kehoe, Kenneth Brower, and Herbert Meier, "U.S. and Soviet Ship Design Practices, 1950–1980," *Proceedings*, Vol. 108/5/951, No. 5, May 1982. On the USN, see Joslyn Fleming, Bradley Martin, Fabian Villalobos, and Emily Yoder, *Naval Logistics in Contested Environments: Examination of Stockpiles and Industrial Base Issues*, RAND Corporation, RR-A1921-1, 2024; Martin et al., 2018.

16 Joel B. Predd, William Kim, and Jay Carroll, *PRC Shipbuilding: Naval and Commercial: A Working Paper Exploring the Relationship Between China's Naval and Commercial Shipbuilding*, RAND Corporation, WR-A2852-1, 2024.

17 Many of these reports and translated articles come from the China Maritime Studies Center based at the U.S. Naval War College. U.S. Naval War College, "China Maritime Studies Institute," webpage, undated.

18 See, for example, Sarah Kirchberger and Christopher P. Carlson, "Neither Fish Nor Fowl: China's Development of a Nuclear Battery AIP Submarine," Center for International Maritime Security, January 22, 2025; Conor M. Kennedy, *China Maritime Report No. 39: A Hundred Men Wielding One Gun: Life, Duty, and Cultural Practices Aboard PLAN Submarines*, China Maritime Studies Institute, 2024; and Roderick Lee, *China Maritime Report No. 27: PLA Navy Submarine Leadership: Factors Affecting Operational Performance*, China Maritime Studies Institute, 2023b.

largely transitioned to a carrier-based force,[19] meaning that the bulk of its remaining aircraft types and units are relatively new—and that there is limited information available for us to study their maintenance practices to date. Both branches, the aircraft carriers themselves,[20] and the rapidly expanding PLAN Marine Corps (PLANMC) deserve future exploration. Finally, we excluded paramilitary maritime forces, such as the China Coast Guard and People's Armed Forces Maritime Militia.

A Brief History of the PLAN from a Strategy and Force-Building Lens

Observers of the USN might identify five periods of peacetime shipbuilding (distinct from wartime mobilization) since the development of the "New Navy" in the 1880s.[21] In thinking about the PLAN, a common saying about modern China is that a generation—whether in terms of people (e.g., foundational experiences growing up) or development (e.g., city skylines)—lasts only five years because of how China's rapid economic development affects its societal and material conditions.[22] A similar refrain can be applied to the PLAN, which has boomed since senior Chinese Communist Party (CCP) and PLA leaders decided to invest in a greater role for the navy during the reform era.[23] These modern epochs can be divided into the phases described in the following sections.

1985 to Early 1990s: Expanding Strategy Greenlights PLAN Modernization

Economic reforms begun in the late 1970s spurred greater interest on the part of CCP leaders in an oceangoing PLAN that could support China's expanding maritime interests, including its growing sea-based trade with the outside world.[24] Senior leaders judged that potential conflicts were not likely to involve total war (such as with the Soviet Union), and, by the late 1980s, increasing emphasis was placed on likely contingencies being local or limited and taking place along China's maritime periphery.[25] Admiral Liu Huaqing, today known as the father of the modern PLAN, additionally helped make the case for an increasing role for the PLAN. By 1985, these developments had (with the approval of Deng Xiaoping, then China's paramount leader) crystallized into the PLAN's first service-specific strategic concept of "near seas active defense."[26] The new concept also set the stage for the growing role of other services compared with the role of the historically ground force–dominant PLA.

Liu's objectives involved building out PLAN presence in the First Island Chain by 2010, the Second Island Chain by 2025, and the Third Island Chain by 2050; this included force planning objectives for fielding

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[19] Rod Lee, *PLA Naval Aviation Reorganization 2023*, China Aerospace Studies Institute, 2023a.

[20] There is clearly interest in U.S. carrier maintenance practices. See for example, Wang Mingwei [王明为] and Tian Xiaochuan [田小川], eds., *U.S. Navy Aircraft Carrier Equipment Maintenance Support Technology and Management* [美国海军航母装备维修保障技术与管理], Harbin Engineering University Press [哈尔滨工程大学出版社], 2017.

[21] Frederick Cichon, "The Next Great Era in U.S. Shipbuilding," *Proceedings*, Vol. 151, No. 2, February 2025.

[22] Alec Ash, "China's New Youth: The Diverse Post-Tiananmen Generation Shaping the Country," China Project, April 8, 2021; John Pomfret, "Exploring the Dreams of China's Pampered and Restless Millennial Generation," *Washington Post*, March 31, 2017.

[23] For a detailed history of this period and factors leading up to the 1985 strategy decision, see Xiaobing Li, *China's New Navy: The Evolution of the PLAN from the People's Revolution to a 21st Century Cold War*, Naval Institute Press, 2023.

[24] Li, 2023, pp. 140–144.

[25] M. Taylor Fravel, *Active Defense: China's Military Strategy Since 1949*, Princeton University Press, 2019, pp. 174–181.

[26] Academy of Military Sciences Strategy Research Department [军事科学院战略研究部], *Science of Military Strategy* [战略学], Academy of Military Sciences Press [军事科学出版社], 2013, p. 208.

aircraft carriers and large, advanced surface combatants by 2000.[27] During the 1980s, PLAN research and development therefore focused on (1) new warship designs featuring more-advanced technology and capable of greater power projection and (2) longer-range missiles and other weapon systems for PLAN assets.[28] However, Deng constrained the military's budget during this period, instead prioritizing economic development. The CCP allowed the PLA to go into private business to make up central government shortfalls to the military, which led to significant corruption within the PLAN and other services. The net result was that few new surface combatants were fielded during these lean budget years because of major delays in shipbuilding programs. For example, only three Luda (Type 051 D/G) ships, the first indigenously designed and produced destroyer class, were commissioned in this era.[29]

Early 1990s to Mid-2000s: A Rising Budgetary Tide Lifts All Boats

After Jiang Zemin became General Secretary of the CCP in mid-1989 following Deng's retirement, Liu continued to rise within the PLA, becoming a member of the top CCP leadership organization, the Politburo Standing Committee, in 1992. Liu's thinking about China's maritime interests and sea power resonated with Jiang.[30] Senior leaders were also laying the policy and legal framework for more-expansive views on China's maritime and territory sovereignty. In 1995, Jiang stated that the PLA should prioritize and accelerate PLAN modernization.[31] Jiang also began the laborious process of divesting the PLA from its commercial activities during this period. As the PLA's overall military strategy codified around the need to fight localized wars under high-technology conditions, leaders assessed that potential conflicts were most likely to be fought in coastal areas and border regions, such as a Taiwan contingency—implying a key role for naval forces.[32]

The PLA as a whole benefited from significant budgetary increases during this period. Key naval force structure trends consisted of select purchases of foreign naval assets, mainly from the newly established Russian Federation, to import advanced naval technology. China acquired four modified *Sovremenny*-class destroyers and Kilo-class diesel submarines.[33] For the domestically built ships constructed during this period, the PLAN continued what one analyst calls the "build a little, test a little" approach of iterative design and production of small ship classes.[34] New destroyers included two ships each in the Luhu (Type 052/052A) and Luyang I (Type 052B) classes, but this approach also applied to the Fuchi (Type 903) class replenishment

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[27] Li, 2023, pp. 153–154.

[28] Li, 2023, p. 154.

[29] These were the now-retired *Guilin* (pennant number 164), *Zhanjiang* (pennant number 165), and *Zhuhai* (pennant number 166), all commissioned between 1987 and 1991. On the significance of this ship class for the PLAN, see *Fighting Ships of the Chinese PLA Navy 1949–2017* [人民海军舰艇拳谱], Modern Navy Magazine Press [《现代舰船》杂志社], undated, p. 148. On commissioning dates, see Richard Sharpe, ed., *Janes Fighting Ships 1990–91*, Jane's Information Group, 1990, p. 109; Janes, "Luda (Type 051D) Class," webpage, November 8, 2019; Janes, "Luda IV (Type 051G II) Class," webpage, September 22, 2020; Andrew Tate, "PLAN Decommissions Four Type 051 Destroyers," Janes, May 17, 2019.

[30] Li, 2023, p. 166.

[31] "Heading Toward the Deep Blue—a Tribute to the People's Navy Celebrating Its 70th Anniversary" ["向着深蓝出发——献给人民海军成立70周年"], Xinhua, April 23, 2019.

[32] Fravel, 2019, p. 209.

[33] Nikolai Novichkov, Yihong Chang, and Richard Scott, "China Buys Two More Project 956EM Ships," *Janes Defence Weekly Naval*, January 8, 2002.

[34] Michael A. McDevitt, *China as a Twenty-First Century Naval Power: Theory, Practice, and Implications*, Naval Institute Press, 2020, pp. 23, 32, 56.

ship.[35] The PLAN also fielded improved electronic countermeasures and radar, sonar, and fire-control systems aboard ships produced during this era.[36]

The PLAN also grappled with maintenance-related challenges during this period. In 2003, a Ming (Type 035) diesel submarine suffered a catastrophic failure in the Bohai Sea, leading to the loss of all lives on board.[37] Maintenance, supply, and training issues were assessed to be the cause, and the accident resulted in the firing of the PLAN commander and political commissar, North Sea Fleet leadership, and the chain of command for submarine maintenance and operations.[38]

Mid-2000s to Mid-2010s: Construction Boom to Enable "Near Seas" Operations

The PLA's overall strategic emphasis evolved during this period from a need to operate under high-technology conditions during the 1990s to an increasing integration and sophistication of military operations under "informatized conditions."[39] By the mid-2000s, official documents stated that the PLA would prioritize PLAN and other service modernization over that of the ground forces.[40] A geographic shift was also underway for the PLAN. In 2004, China's national defense white papers began consistently emphasizing that the PLAN was expanding its operations into "offshore" or "near seas" [近海] areas, which had been the navy's strategic concept since the mid-1980s.[41]

Increasingly during this era, PLAN operational focus areas had both warfighting and broader international remits.[42] In 2004, new Central Military Commission chairman Hu Jintao issued objectives for the PLA known as the "new historic mission and tasks," which called on the PLA to actively support China's development.[43] This guidance stipulated a need for the PLA to defend China's interests in emerging domains—

[35] McDevitt, 2020, pp. 21–27, 31, 56.

[36] Li, 2023, p. 169.

[37] Bernard D. Cole, "The Evolution of China's Naval Strategy," National Bureau of Asian Research, March 26, 2024.

[38] Cole, 2024; Jeffrey Becker, David Liebenberg, and Peter Mackenzie, *Behind the Periscope: Leadership in China's Navy*, CNA, 2013, p. 81. See also Lyle Goldstein and William Murray, "Undersea Dragons: China's Maturing Submarine Force," *International Security*, Vol. 28, No. 4, Spring 2004, p. 164.

[39] Fravel, 2019, p. 219.

[40] State Council Information Office of the People's Republic of China, *China's National Defense in 2004* [2004年中国的国防], December 2004a; State Council Information Office of the People's Republic of China, *China's National Defense in 2004*, trans., December 2004b.

[41] State Council Information Office of the People's Republic of China, *China's National Defense in 2008* [2008年中国的国防], January 2009a; State Council Information Office of the People's Republic of China, *China's National Defense in 2008*, trans., January 2009b. The 2008 white paper states,

> From the 1950s to the end of the 1970s the main task of the Navy was to conduct inshore [or coastal] defensive operations. Since the 1980s, the Navy has realized a strategic transformation to offshore [or near seas] defensive operations. Entering the new century, in view of the characteristics and patterns of local maritime wars under conditions of informatization, the Navy has been striving to all-around improve its capabilities of integrated offshore [or near seas] operations.

See also State Council Information Office of the People's Republic of China, 2004a; State Council Information Office of the People's Republic of China, 2004b; State Council Information Office of the People's Republic of China, *China's National Defense in 2006* [2006年中国的国防], December 2006a; and State Council Information Office of the People's Republic of China, *China's National Defense in 2006*, trans., December 2006b.

[42] The two roles were not necessarily separate in the minds of PRC leaders. The Office of Naval Intelligence noted in a 2009 report that "when Hu first called for this capability in his 2004 expansion of the historic missions, concerns over the Malacca Strait dominated discussion of [sea line of communication] security." Office of Naval Intelligence, *The People's Liberation Army Navy: A Modern Navy with Chinese Characteristics*, August 2009, p. 11.

[43] James Mulvenon, "Chairman Hu and the PLA's 'New Historic Missions,'" *China Leadership Monitor*, No. 27, January 2009; Cortez A. Cooper, "The PLA Navy's 'New Historic Missions': Expanding Capabilities for a Re-Emergent Maritime Power," tes-

including the maritime domain, which played an increasingly important role in China's economic development.[44] To protect China's overseas interests, the PLA would need to operate increasingly far from China's shores; the PLAN's participation in United Nations–authorized counterpiracy patrols in the Gulf of Aden beginning in 2008 can be viewed as one output of this direction.[45]

Naval shipbuilding experienced rapid growth during this period. As a 2024 RAND working paper notes, PLAN shipbuilding had an average of "250,000 active tonnage under construction . . . [from 2003 to 2009]. That number doubled in 2010, reaching a peak in 2014 of over 800,000 active tonnes."[46] By the mid-2000s, the PLAN had largely begun shifting away from the "build a little, test a little" approach in favor of series production of larger ship classes. The PLAN commissioned six Type 052C DDGs (guided missile destroyers) between 2005 and 2015, and then commissioned its first Type 052D ship in 2014, a class—now at more than three dozen ships and counting—that was still being produced as of 2025.[47] A similar transition within the frigate force occurred with the beginning of series production of the Jiangkai II (Type 054A). Series production could theoretically help the PLAN streamline maintenance requirements and processes and associated training. However, as discussed later in this report, the increased production and fielding of large surface combatants during this period might have begun to stress the maintenance workforce and associated organizations, such as vessel training centers (VTCs), which ensure ship deployability.

Mid-2010s–Present: Increasing Specialization amid Expansion to "Near Seas Defense and Far Seas Protection"

The boundary between this era and the previous one is admittedly fuzzy, given that Hu began calling for China to develop into a "maritime great power" during the end of his tenure in 2012.[48] Admiral (Ret.) Michael McDevitt notes that this was not merely the last word from a leader departing the scene but a judgment reached in the 18th Party Congress Work Report, a highly authoritative document indicating the CCP's key priorities and assessments of progress made over the five years preceding the report.[49] After Xi Jinping assumed the top leadership billet in 2012, emphasis on strengthening China's maritime and naval capability continued.[50] From an interservice and jointness perspective, the 2015 national defense white paper drew a

timony before the U.S.-China Economic and Security Review Commission, RAND Corporation, CT-332, June 11, 2009. The original concept was later revised to (1) "provide strategic support for consolidating the leadership position of the CCP and the socialist system"; (2) "provide strategic support for the safeguarding of national sovereignty, unity, and territorial integrity"; (3) "provide strategic support for protecting China's overseas interests"; and (4) "provide strategic support for promoting world peace and development." See also Yan Wenhu [闫文虎], "Correctly Understand the Military's Missions in the New Era" ["正确理解新时代军队使命任务"], *China Military Online* [中国军网], July 26, 2019.

[44] Fravel, 2019, p. 229.

[45] Dennis J. Blasko, "CMSI Note #8: Recent Changes in the PLA Navy's Gulf of Aden Deployment Pattern," China Maritime Studies Institute, 2024.

[46] Predd, Kim, and Carroll, 2024.

[47] There are three subvariants within the overall class, but the main design is the same: the Type 052D variant; the Type 052DL variant, which is 4 m longer to accommodate larger helicopters; and the Type 052DM variant with improved radars. See Janes, "Luyang III (Type 052D/052DL/052DM) Class (DESTROYERS) (DDGHM)," webpage, January 22, 2026a.

[48] McDevitt, 2020, p. 1.

[49] McDevitt, 2020, p. 1.

[50] Christopher H. Sharman and Andrew S. Erickson, "China's Future World-Class Navy: Ends, Ways, and Means," in Benjamin Frohman and Jeremy Rausch, eds., *The PLA's Long March Toward a World-Class Military: Progress, Obstacles, and Ambitions*, National Bureau of Asian Research, 2025, p. 202. See also Wu Yue [伍岳], Chen Yan [陈杨], and Li Liang [李亮], "Thoughts on the Construction of Modern Military Logistics System of the Navy" ["海军现代军事物流体系建设问题思考"], *World Navy Research* [世界海军研究], No. 3, 2023, p. 72.

line in the sand by stating that "the traditional mentality that land outweighs the sea must be abandoned," foreshadowing PLA-wide changes in 2015–2016 that placed the ground forces on an equal organizational footing with the other services.[51]

Also in 2015, the PLAN formally expanded its strategic concept from "near seas defense" to "near seas defense and far seas protection," signifying that greater roles and geographic reach were becoming the expectation for future naval operations.[52] PLAN force structure developments reflect a continued push to grow the fleet in terms of numbers and capability. Although overall active naval tonnage under construction dropped after 2014, series production of major combatants continued.[53] More than one-half of the ships participating in a 2018 PLA naval review were commissioned during Xi's tenure,[54] indicating the relatively young age of the PLAN's most-advanced surface assets. New large surface combatants commissioned between when Xi took power in late 2012 and January 2026 include at least nine Renhai (Type 055) cruisers and 34 Luyang III (Type 052D) destroyers,[55] as well as frigates, corvettes, and replenishment vessels.[56] As explored later in this report, some sources we reviewed refer to the PLAN rapidly commissioning numerous ships "like dumplings [one after the other]" during this period, raising the demand for maintenance and other activities.[57]

On the infrastructure side, naval shipbuilding became more specialized during this period, in contrast to the earlier mix of commercial and naval orders being fulfilled at various shipyards. This is likely because of the increasing technological complexity of PLAN ships and the demand for specific workforce expertise that this complexity implies. Specific shipyards (mainly Dalian and Jiangnan Changxingdao) produce the PLAN's most-advanced surface combatants.[58]

Looking ahead, what are leaders' expectations for China's navy? And how might those expectations inform ongoing and future force modernization? Mirroring what he required for the other PLA services, Xi called on the PLAN in 2018 to become a "world-class navy."[59] Although neither Xi nor PLAN leaders have clearly stated their requirements for achieving world-class naval status, one 2024 assessment noted that "PLA writings suggest this 'world-class' goal is tied to both technological development and the navy's capability

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[51] State Council Information Office of the People's Republic of China, *China's Military Strategy* [中国的军事战略], May 2015a; State Council Information Office of the People's Republic of China, *China's Military Strategy*, trans., May 2015b.

[52] Jennifer Rice and Erik Robb, *China Maritime Report No. 13: The Origins of "Near Seas Defense and Far Seas Protection,"* China Maritime Studies Institute, 2021. This transition had been underway since at least 2004. See Academy of Military Sciences Strategy Research Department, 2013, p. 209.

[53] Predd, Kim, and Carroll, 2024, p. 3.

[54] Sharman and Erickson, 2025, p. 198.

[55] Although the PLAN designates the Renhai ship class as a destroyer [驱逐舰], U.S. and other sources refer to it as a cruiser because of its size, displacement, and quantity of vertical launch system (VLS) cells. Thanks to Josh Arostegui for raising this point.

[56] Quantities are based on information from Janes updated as of January 26, 2026. The appendix includes CVM (aircraft carrier), CG (guided missile cruiser), and DDG commissioning dates. One Renhai CG is listed as commissioning sometime in 2026, which would be the tenth commissioned since Xi took power. In our Luyang III count, we include one ship for which the commissioning date was not available, but the ships launched before and after it have both already commissioned; another Luyang III DDG is listed as commissioning sometime in 2026, which would be the 35th commissioned since Xi took power. For details on other ship classes, see Predd, Kim, and Carroll, 2024.

[57] Duan Jiangshan [段江山], "Huangshan Warship Sails Far Away, a New Answer to Equipment Support" ["黄山舰远航，装备保障的一张新答卷"], *PLA Daily* [解放军报], October 19, 2018.

[58] Predd, Kim, and Carroll, 2024, p. 7.

[59] On the topic of "world-class military" and analysis for each PLA service, see Benjamin Frohman and Jeremy Rausch, eds., *The PLA's Long March Toward a World-Class Military: Progress, Obstacles, and Ambitions*, National Bureau of Asian Research, 2025.

to execute specified missions," including the ability to achieve sea control within the First Island Chain and successfully contest the USN in establishing sea control.[60] Since 2019, another layer was added to the PLAN's strategic concept to include not only "near seas defense and far seas protection" but also "oceanic presence, and expansion into the two poles."[61]

In terms of innovating to become world-class, a 2017 speech by then–PLAN commander Admiral Shen Jinlong highlighted the need to achieve intelligentized warfare by pursuing advancements in such applications as the Internet of Things and smart manufacturing, both of which could augment naval logistics.[62] Chapter 3 further discusses PLAN interest in these types of technologies. A 2023 article in the PLA journal *World Navy Research* notes that the increasingly complex strategic environment China faces requires improvements to its naval logistics and supply chains.[63] The authors state that, with the "strong enemy" of the United States continuing to encircle China and strengthen its own capabilities, "the navy's construction of a modern military supply system [物流体系] directly confronts the front lines of our maritime borders and is crucial to the reach of our national military power."[64] The article goes on to explain the need to "strengthen systematic research based on logistics [后勤] theory guidance, establish authoritative regulations through top-level design, clarify conceptual meanings, define essential boundary factors, and distinguish between 'logistics' and 'supply chain management.'"[65]

Introduction to PLAN Maintenance

Maintenance practices in the PLAN broadly mirror those of the USN. Maintenance is categorized into various levels of complexity and sophistication and conducted by a mix of onboard maintainers and broader enterprise personnel. However, changes in the PLAN's organizational structure since the 2015–2016 PLA-wide reforms have had potentially important implications for ship maintenance by likely enabling PLAN maintenance activities to become more flexible at lower levels. Advantages of these organizational changes could involve better integrating PLAN maintenance activities into theater-specific and joint operations while maintaining the benefits of centralizing certain aspects of maintenance execution and supporting activities, such as standardizing maintenance requirements.

PLAN Headquarters

As part of the PLA's major organizational overhaul declared in early 2016, operational control of many naval units moved from PLAN Headquarters (PLAN HQ) to the naval component commands of the PLA's Theater

[60] Sharman and Erickson, 2025, pp. 203–204.

[61] Sharman and Erickson, 2025, p. 202.

[62] Sharman and Erickson, 2025, p. 205. On the term *intelligentization*, see Office of the Secretary of Defense (OSD), *Annual Report to Congress: Military and Security Developments Involving the People's Republic of China 2025*, U.S. Department of Defense, 2025, p. 63.

[63] Wu, Chen, and Li, 2023.

[64] Wu, Chen, and Li, 2023, p. 72. This article discusses both supply and logistics [后勤] systems. The term *supply system* [物流体系] could be translated as *logistics system* in a broader sense, such as a business or commercial context, and is how the authors of the article translated the term into English, but this is not the typical PLA term for logistics [后勤]. The 2011 PLA dictionary defines 军事物流 as "military supply flow." All-Military Military Terminology Management Committee [全军军事术语管理委员会], Chinese People's Liberation Army Military Terminology [中国人民解放军军语], Academy of Military Sciences Press[军事科学出版社], 2011, p. 497.

[65] Wu, Chen, and Li, 2023, p. 74.

Commands.[66] The arrangement is more similar to the U.S. military's command structure, in which commanders of regionally defined unified combatant commands have operational control over a given force while service heads have administrative control. However, PLAN HQ appears to retain control over certain forces, such as the PLANMC and the service's aircraft carriers and the PLAN's small number of overseas bases.[67]

The role of PLAN HQ as it pertains to maintenance primarily lies within its Equipment Department, which probably seeks to gain efficiencies and uniformity in surface ship maintenance by centralizing and standardizing components of maintenance, such as training standards, technical specifics of parts, and major maintenance schedules, similar to USN system commands.[68] Brief descriptions of the roles and responsibilities of each of the four major administrative and functional departments in protocol order, including information on subordinate organizations relevant for PLAN maintenance, are as follows:[69]

- The **Staff Department** is responsible for a variety of administrative and functional departments, including operations and training. This department manages the Training Bureau and PLAN academic institutions, which provide technical, tactical, and leadership education to PLAN officers and noncommissioned officers (NCOs), presumably including maintenance training and a host of other training topics.
- The **Political Work Department** manages CCP functions; operational functions, such as public opinion, psychology, and law; and administrative functions regarding personnel management, officer selection, and professional military education. The Political Work Department likely has minimal impact on maintenance outside the important role of personnel management, although the broader CCP apparatus within the PLAN, such as unit Party Committees and political commissars, is dissimilar to that of the USN.
- The **Logistics Department** is responsible for the financing, acquisition, and management of facilities, supplies, and other services. Its role in procurement and supply management is important in ensuring that repair equipment and spare parts remain available. The department also can coordinate with the Central Military Commission's Logistics Support Department, the Joint Logistic Support Force, and the veteran affairs ministry, which is responsible for a network of local government-run military supply facilities.
- The **Equipment Department** oversees major PLAN platforms and weapon programs, including surface vessels. Critically, this department likely oversees programmatic planning, the scheduling and synchronization of major maintenance periods across fleets, and the development of maintenance requirements for ship classes.[70] Additionally, it manages the network of Military Representative Bureaus that likely conduct quality control of PLAN equipment throughout the acquisition process.[71] The bureaus have both research institutes and subordinate offices that engage the defense industry at the factory level.

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66 Joel Wuthnow and Phillip C. Saunders, *Chinese Military Reforms in the Age of Xi Jinping: Drivers, Challenges, and Implications*, Institute for National Strategic Studies, National Defense University, 2017. Note there are only three Theater Command Navies (TCNs)—the Western and Central Theater Commands do not have naval component commands.

67 Brian Waidelich, "People's Liberation Army Navy," in Frank Miller, Tung Ho, and Kenneth W. Allen, eds., *The People's Liberation Army as Organization*, Vol. 3.0, Jamestown Foundation, 2025, p. 148.

68 Some subordinate departments within PLAN HQ first-level departments might have had their functions moved to TCNs as part of broader organizational changes. See Kevin Pollpeter and Kenneth W. Allen, eds., *The PLA as Organization*, Vol. 2.0, Defense Group Incorporated, 2012, p. 308.

69 Descriptions largely adapted from Waidelich, 2025, pp. 148–150.

70 Peter A. Dutton and Ryan D. Martinson, *China's Evolving Surface Fleet*, China Maritime Studies Institute, July 2017.

71 Waidelich notes, "It appears that this arrangement of military liaison offices within the PRC's defense industry is not reciprocated by the civilian side. That is, there do not appear to be formal organizational structures in place for detailing PRC defense contractor representatives to PLA naval commands" (Waidelich, 2025, p. 150). For a history of the PLAN HQ Equipment Department and subordinate-level Support Departments, see Kenneth W. Allen, *PLA Navy Equipment Department and Support Department: History and Organizational Background*, China Analysis from Original Sources website, January 22, 2025, pp. 2–3.

Theater Command Navies

The three TCNs are the naval component commands of the Northern, Eastern, and Southern Theater Commands. The Theater Commands have operational control over the TCNs; PLAN HQ has administrative control.[72] This is a change from the previous system, in which the regional commands established operational command over the regional navies only during wartime. Given that TCNs have operational control over forces from multiple branches (e.g., subsurface, surface, naval aviation, and coastal defense units), it is possible that maintenance activities are now more closely coordinated across branches (e.g., ensuring that vessels are available to participate in support of an exercise conducted by shore-based aircraft). Table 1.1 provides an overview of surface combatants by TCN.

Each TCN comprises three departments, one fewer than at the PLAN HQ level. The TCN's Support Department appears to fill similar roles to PLAN HQ's Logistics and Equipment Departments and likely plays a central role in planning maintenance and organizing related material procurement. Each TCN also has a VTC that provides basic training for crews of newly commissioned ships and ships that have completed major upgrades; the VTC also evaluates ships for basic readiness and individual sailors against training objectives.[73] These mostly parallel structures and roles across TCNs likely create further efficiencies in interactions with PLAN HQ and within a TCN.

Within each TCN, Support Departments or organizations with equivalent roles and responsibilities exist at various echelons, including the flotilla or detachment [支队; *zhidui*] level.[74] These are shore-based organizations with a variety of maintenance-related roles, including mobility, supply, and inspection functions.[75]

Maintenance Levels and Types

The PLAN appears to organize surface ship maintenance periods into a system similar to the USN's approach for its surface fleet.[76] Routine maintenance while underway relies primarily on onboard sailors augmented as needed (either in person or, increasingly, virtually) by technical personnel from the shore-based organizations described earlier. These shore-based organizations perform higher-complexity maintenance while a ship is in port, typically at the flotilla level, although there is some evidence to suggest that a portion of these

[72] Waidelich, 2025, p. 158.

[73] Waidelich, 2025, p. 160.

[74] For surface ships, subordinate organizations include the *flotilla*, which is sometimes called a *detachment* [支队; *zhidui*]; the *squadron* [大队; *dadui*]; and the *division* [中队; *zhongdui*]. We adopt the translations of (1) *squadron* for *dadui* because of common use by Western analysts of the PLA and (2) *division* for *zhongdui* based on historical USN use of this term for similar-sized units. See use of *squadron* in Waidelich, 2025, pp. 175–179; Pollpeter and Allen, 2012, pp. 52–53; and Tiffany A. Tat, "CMSI Note #7: PLA Navy Reserve: Out of the Shadows and into the Forefront?" China Maritime Studies Institute, August 5, 2024.

[75] Fan Jianghuai [范江怀], Li Wei [李维], and Zhang Yi [张毅], "Regarding the Debate of Understanding 'What Happens Behind the Scenes'" ["关于"台前幕后"的认知之辩"], *PLA Daily* [解放军报], September 7, 2021. Although the organization of maintenance and other sustainment functions below the theater level is not fully understood, *dadui*/squadrons likely have subordinate units responsible for a variety of systems, such as missiles, ordnance, fuels, and naval vessel equipment.

[76] The USN organizes maintenance activities into three categories: organizational maintenance, performed by crews as part of their duties while underway or in port; intermediate maintenance, which exceeds the capacity of the crew and requires additional support, such as the use of fleet maintenance organizations and can include such tasks as the calibration, repair, refurbishment or replacement of damaged or unserviceable parts; and Chief of Naval Operations (also called *depot*) maintenance, which exceeds the capacity of an intermediate maintenance facility and can be performed at a public or private shipyard, typically involving major repairs and overhauls. Dutton and Martinson, 2017; and GAO, *Navy Ship Maintenance: Actions Needed to Monitor and Address the Performance of Intermediate Maintenance Periods*, GAO-22-104510, February 2022a.

TABLE 1.1
Surface Combatant Organization, by Theater Command Navy

TCN	Flotilla Number and Type (Location)	Major Modern Surface Combatants
Northern	• 1st Destroyer Flotilla (Qingdao) • 10th Destroyer Flotilla (Dalian) • 11th Frigate Flotilla (Dalian) • 12th Frigate Flotilla (Qingdao)	• 11 Type 052D/DL/DM Luyang III DDGHMs • 1 Type 054B Jiangkai III FFGHMs • 11 Type 054A/AG Jiangkai II FFGHMs • 4 Type 055 Renhai CGHMs
Eastern	• 3rd Destroyer Flotilla (Zhoushan) • 6th Destroyer Flotilla (Zhoushan) • 13th Frigate Flotilla (Lianyungang) • 14th Frigate Flotilla (Shanghai) • 15th Frigate Flotilla (Ningde) • 16th Frigate Flotilla (Xiamen)	• 11 Type 052D/DL/DM Luyang III DDGHMs • 19 Type 054A/AG Jiangkai II FFGHMs • 2 Type 055 Renhai CGHMs
Southern	• 2nd Destroyer Flotilla (Zhanjiang) • 9th Destroyer Flotilla (Sanya) • 17th Frigate Flotilla (Guangzhou) • 18th Frigate Flotilla (Beihai) • 19th Frigate Flotilla (Sanya)	• 13 Type 052D/DL/DM Luyang III DDGHMs • 1 Type 054B Jiangkai III FFGHM • 14 Type 054A/AG Jiangkai II FFGHMs • 4 Type 055 Renhai CGHMs

SOURCES: Features information from Waidelich, 2025; Janes, "China—Navy," webpage, September 1, 2025c; Janes, 2026a; Janes, "Jiangkai III (Type 054B) Class (FFGHM)," webpage, January 28, 2026d; Janes, "Jiangkai II/Improved Jiangkai II (Type 054A/054AG) Class (Frigates) (FFGHM)," webpage, January 23, 2026b; Janes, "Renhai (Type 055) Class (CRUISERS) (CGHM)," webpage, January 28, 2026e; Janes, "Fujian (Type 003) Class (CVM)," webpage, January 26, 2026c; Janes, "Type 001 and Type 002 (Modified Kuznetsov/Orel) (Modified Project 1143.6) Classes (CVM)," webpage, July 16, 2025b; Janes, "China Commissions Fujian Aircraft Carrier," *Janes Defence Weekly*, November 7, 2025f; Chris Buckley and Amy Chang Chien, "China's New Aircraft Carrier Signals Naval Ambitions," *New York Times*, November 7, 2025.

NOTE: CGHM = cruiser with surface-to-surface missiles, a hangar, and surface-to-air missiles; DDGHM = destroyer with surface-to-surface missiles, a hangar, and surface-to-air missiles; FFGHM = frigate with surface-to-surface missiles, a hangar, and surface-to-air missiles. The surface combatants listed in this table represent the PLAN's most-modern destroyers and frigates commissioned and assigned to a TCN, according to information updated as of January 26, 2026, and are not a complete picture of the vessels in each TCN. One of the Renhai CGs assigned to the Eastern TCN is expected to commission sometime in 2026. It is unclear whether the PLAN's aircraft carriers are formally assigned to TCNs, given that they are subordinate to PLAN HQ. *Liaoning* (pennant number 16), an overhauled *Kuznetsov*-class carrier, was commissioned in 2012 and typically operates with the Northern TCN. *Shandong* (pennant number 17), a modified *Kuznetsov*-class design, was commissioned in December 2019 and typically operates with the Southern TCN. *Fujian* (pennant number 18) is a novel design exceeding 80,000 tons displaced and featuring electromagnetic catapults. *Fujian* was commissioned in November 2025 at Sanya Naval Base, potentially signaling an affiliation with the Southern TCN.

repairs are contracted to civilian maintainers. For major repairs and upgrades, the PLAN appears to rely on a mix of Support Department organizations and civilian contractors.[77]

According to a 2022 PLA journal article, PLAN ship maintenance is classified into two main types: *unplanned maintenance* [临时维修], or "non-preventive repair of ship failures, damage, and battle damage," and *planned maintenance* [计划维修], which is "a regular task of dismantling and repairing ships at regular intervals according to their service life."[78] Repairs are divided into four levels—dock, minor, medium, and mid-life or major:

> Dock repair, also known as dock maintenance, is the regular entry of ships into the dock for inspection and maintenance. It can also be carried out on the slipway. The purpose is to remove debris and rust from the underwater part of the ship and to maintain and repair mechanical and electrical equipment.
>
> Minor repairs are local preventive dismantling and repairs of the hull and various equipment after the ship has been used for a certain number of years. The purpose is to maintain it in normal technical condition before the next repair.

[77] RAND analysis of 829 maintenance contract–related procurement documents found 19 documents for "ship-grade repair" contracts (an intermediate level of maintenance) and 11 documents for "fleet-grade repair" contracts. In these instances, an element of the Eastern TCN solicited and awarded a series of contracts in multiple tranches for private companies to execute various maintenance operations.

[78] Ju Lang, *CMSI Translations #11: Mid-Life Overhaul and Upgrade of the Type 052C Guided Missile Destroyer*, China Maritime Studies Institute, December 19, 2024, p. 11. The article was translated in 2024 but was originally published in 2022.

> Mid-life overhaul is a comprehensive preventive dismantling and repair of a ship to restore it to its original state and maintain its original performance. The scope and workload of the mid-life overhaul are larger, and the implementation period is also longer.
>
> A major overhaul is a comprehensive inspection and repair of a ship to fully restore its original performance. However, due to the characteristics of mechanical and electronic equipment, this is almost impossible to achieve, so usually only two mid-life overhauls are carried out during the life of the ship.[79]

Medium repair is not explained in the 2022 article, but a PLAN maintenance dictionary states, "This refers to a relatively comprehensive dismantling, inspection, and repair of a warship after several dockings (or berthings) and minor repairs. The purpose is to restore the warship's basic tactical and technical capabilities."[80]

Understanding Maintenance Activities at Echelon

Table 1.2 summarizes naval maintenance activities listed in the PLA sources we reviewed, along with the echelon at which we assessed that these activities occurred.[81]

Organization of This Report

The rest of this report is organized into three chapters. In Chapter 2, we examine the relationship between force design, shipbuilding, and maintenance, applying insights across maintenance archetypes to understand trends applicable to the PLAN's rapid modernization. In Chapter 3, we synthesize key themes we identified in the PLAN's evolving approach to maintenance management based on a review of PLA sources. In Chapter 4, we build on this analysis and offer findings and implications. The appendix lists the launch and commissioning dates of the PLAN's large surface combatants (aircraft carriers, cruisers, and destroyers) since 2000.

[79] Ju Lang, 2024, p. 11.

[80] Navy Ship Equipment Maintenance Theory Research Editorial Office [舰船装备维修理论研究室编], *Terminology of Navy Ship Equipment Maintenance* [舰船装备维修名词术语], Navy Equipment Maintenance Bureau Press [海军装备修理部出版], 1991, p. 5. This dictionary lists a definition for "major repair" similar to the overhaul definition provided earlier and similar definitions for "dock repair" and "minor repair."

[81] Because of limitations in determining the exact number, composition, and capabilities of PLAN maintenance units at lower echelons, this table provides examples of the types of activities found at echelon to inform our analyses. A more complete understanding of PLAN maintenance organization force structure is an area for further research.

TABLE 1.2

PLAN Maintenance Activities at Echelon

Assessed Echelon of Activity (Example Institution)	Example Activities (with TCN If Mentioned)	Specific Echelon or Other Details
Central Military Commission department (e.g., Equipment Development Department [EDD] [装备发展部])	The EDD and the Ministry of Human Resources and Social Security organize an annual equipment maintenance skills competition; 354 PLA and military state-owned enterprise personnel competed in it in 2021, including PLAN and shipbuilding industry maintainers.[a]	No additional details provided
Theater-level naval component command (e.g., Eastern TCN [东战区海军])	A TCN sends a fleet delegation to academic institutions to exchange information on equipment support (Southern TCN to Naval University of Engineering).[b]	Southern TCN leaders lead delegation
	A TCN organizes a Theater Command naval professional skills competition, won by an electromechanical technician one year.[c]	No additional details provided
Shore service departments [岸勤部] at naval bases [海军基地], including Comprehensive Support Bases and other support bases (e.g., Zhoushan Comprehensive Support Base [舟山综合保障基地]	A naval base creates a modular "equipment support and maintenance team" [装备支援维修组] and other modular teams to participate in a "comprehensive support training" activity.[d]	Base leadership discusses plans to continue improving comprehensive support; training might have occurred pierside, although no details are provided
	A naval base organizes maintenance training events and competitions, including an equipment support drill[e] and a mechanical welding competition (Northern TCN).[f]	Unclear whether the relevant flotilla(s) had some involvement in organizing the exercises
Note: The repair facilities and service support vessel division mentioned in these example activities are not at this echelon but are subordinate to the base. The training base is subordinate to the TCN.	A repair facility checks the maintenance period of ship-based equipment (e.g., life raft) and conducts repairs prior to long-distance patrol (Southern TCN).[g]	Checks led by a senior engineer at a repair facility [某修理厂高级工程师]
	A repair facility provides remote guidance to a ship crewmember to resolve abnormal equipment status (Southern TCN).[g]	No additional details provided
	A repair facility prepares maintenance teaching materials and video tutorials and coordinates with the manufacturer's technical personnel and academic experts to conduct skills training as a way to improve organic crew familiarity with less commonly used equipment (Southern TCN).[g]	Manufacturer's technical personnel [厂家技术人员]
	A service support vessel division [勤务船中队] carries out equipment inspections of ships (Northern TCN).[h]	Division instructor encourages personnel not to expect remote support from manufacturers' technicians [厂家师傅] while on the battlefield
	A naval training base coordinates with vocational schools to align their curricula for future NCOs with the training base's needs.[i]	Naval training base officials and personnel coordinate with college staff
	Nonspecialized personnel at a comprehensive support base train for such secondary roles as plumbers and electricians and conduct minor repairs at the base (e.g., dormitory water heaters) (Northern TCN).[j]	Personnel are pierside

Table 1.2—Continued

Assessed Echelon of Activity (Example Institution)	Example Activities (with TCN If Mentioned)	Specific Echelon or Other Details
Port-level activities on the water (e.g., within the surface ship flotilla [支队] and/or in coordination with the TCN Support Department [战区海军保障部])	A flotilla's leadership approves extending a component's maintenance cycle, following a submarine's "major overhaul coordination meeting" [大修协调会上].[k]	Anecdote suggests flotilla leadership would have authority to approve or adjust overhaul planning
	A flotilla manages maintenance schedules to optimize for readiness, balancing maintenance needs and training cycles (Southern TCN).[l]	Flotilla Party Committee worked with its shipyard [船厂]
Note: The example activities here could involve organizations that are not at this echelon but are subordinate to the flotilla and/or TCN Support Department.	While accompanying a ship during a mission, a senior NCO from a naval technical support *dadui*/squadron [海军某技术保障大队], repairs a deck mooring bollard after organic maintenance personnel are unable to make the repair.[m]	Example of an enterprise technician accompanying a ship during operations
	A military academic consults on a ship's electrical engineering after crew is "unable to independently eliminate" a fault (Eastern TCN and Naval University of Engineering).[b]	Unclear whether the flotilla or ship leadership issued the invitation
	A unit sends an NCO to lecture at military academy; the NCO compiles teaching materials for use by the academy and units (Eastern TCN to Naval Submarine Academy).[k]	Echelon unclear but flotilla and/or submarine leadership likely approved request for the NCO to teach
	An electromechanical technician wins first place in the flotilla's professional skills competition two years in a row.[c]	Flotilla organized the competition; winner is a corporal
Onboard maintenance (e.g., conducted while underway by organic maintenance personnel [often 部队技术人员 or 部队专业骨干])	Ship crew conducts an emergency diesel engine repair after temperature abnormality (Southern TCN).[n]	No additional details provided
	Submarine crew members repair a pipeline (both examples in notes from Northern TCN).[o, p]	No additional details provided
	During a "comprehensive support training" activity, the ship's crew is remotely supported by an offsite "technical team" [技术团队] via video feed to complete repairs.[d, q]	Training activity might have occurred pierside; no details are provided
	Ship crew repairs a central system display and control console with help from the onboard manufacturer's technical personnel [厂家随舰技术人员] (Southern TCN).[n]	Repair conducted the night before the ship departed for a mission (i.e., conducted pierside)
	A submarine crew member conducts emergency troubleshooting of a power unit and eliminates the fault (Northern TCN).[p]	No additional details provided
	A ship crew member inspects pipelines.[c]	No additional details provided

Table 1.2—Continued

Assessed Echelon of Activity (Example Institution)	Example Activities (with TCN If Mentioned)	Specific Echelon or Other Details
(Continued) Onboard maintenance (e.g., conducted while underway by organic maintenance personnel [often 部队技术人员 or 部队专业骨干])	A patrol boat unit provides information to manufacturers to improve knowledge sharing (Southern TCN).[r]	Unit is based in the Spratlys
	Other equipment types with examples of repairs are radar equipment (Southern TCN example),[s] main guns (Southern TCN),[t] sonar equipment,[u] and communications equipment (Eastern TCN).[v]	Sonar example involved a telephone consultation from a remote technician

SOURCES: Features information on echelon relationships from Rick Gunnell, "Central Military Commission," in Frank Miller, Tung Ho, and Kenneth W. Allen, eds., *The People's Liberation Army as Organization*, Vol. 3.0, Jamestown Foundation, 2025; Waidelich, 2025; Nan Li, "Chapter Nine: The People's Liberation Army Navy as an Evolving Organization," in Kevin Pollpeter and Kenneth W. Allen, eds., *The PLA as Organization*, Vol. 2.0, Defense Group Inc., 2012, p. 327. Information on shore service departments from Kennedy, 2026, pp. 98–102.

[a] "'Craftsmanship Cup' Equipment Maintenance Vocational Skills Competition Opens" [""匠心杯" 装备维修职业技能大赛开幕"], *PLA Daily* [解放军报], September 26, 2021.

[b] Xiong Feng [熊峰], "Topics Originate from the Troops; Results Go to the Troops" ["课题从部队中来 成果到部队中去"], *PLA Daily* [解放军报], October 12, 2021.

[c] Jiang Haitao [姜海涛], Wang Desai [王德赛], and Jia Siyu [贾思宇], "Navy Electromechanical Troops: Although the Post in the Cabin Is Small, the Responsibility to Protect the Long Voyage Is Heavy" ["海军机电兵：岗位虽小在舱底，责任重大护远航"], *PLA Daily* [解放军报], August 16, 2024.

[d] Feng Guangming [冯光明道] and Zhang Yi [张毅报], "Coordinating Key Elements to Raise the Level of Comprehensive Support" ["要素协同提升综合保障水平"], *PLA Daily* [解放军报], December 12, 2014.

[e] There are three broad levels of PLA training: "A *basic training* (训练) activity includes basic day-to-day training on an individual, subunit, or unit level, generally on one training subject. *Drills* (演练) involve multiple training subjects. An *exercise* (演习) usually occurs at the regiment or brigade level and above and can involve one or more units. It can last a day or more and incorporates the most sophisticated training." Bonny Lin and Cristina L. Garafola, *Training the People's Liberation Army Air Force Surface-to-Air Missile (SAM) Forces*, RAND Corporation, RR-1414-AF, 2016, pp. 16–17.

[f] Zhou Jianlong [周建龙], Cheng Yu [程禹], and Sui Yifan [隋一范], "Maintenance Tools Seem to 'Grow' on His Hands" ["维修工具像"长" 在手上"], *PLA Daily* [解放军报], May 19, 2023.

[g] Li Lingjie [李玲杰] and Zhang Yi [张毅], "Combining Training and Maintenance to Help Warships Sail Far Away" ["修训结合助力战舰远航"], *PLA Daily* [解放军报], July 22, 2023.

[h] Zhou Jianlong [周建龙] and Xu Chao [许超], "The Equipment Gets Used and Must Be Repairable" ["装备会用也要能维修"], *PLA Daily* [解放军报], October 28, 2021.

[i] Wu Qiang [吴强] and Liu Mingkui [刘明奎], "At the Starting Line; Running the First Leg Well" ["发力起跑线 跑好第一棒"], *PLA Daily* [解放军报], June 12, 2024.

[j] Cui Xu [崔旭], Li Yinchuan [李银川], and Wu Xingyu [邬兴羽], "Let's Praise Our 'Five Small Workers'" ["夸夸咱们的"五小工""], *PLA Daily* [解放军报], May 29, 2021.

[k] Fan Jianghuai [范江怀], Liu Yaxun [刘亚迅], and Dai Zongfeng [代宗锋], "Deep Sea Assault" ["深海突击"], *PLA Daily* [解放军报], December 16, 2018.

[l] Wu Fangbing [吴方兵] and Chen Dianhong [陈典宏], "The Ships Are Being Repaired at the Yard; the Soldiers Go to Sea for Training" ["船在厂里修 兵赴海上练"], *PLA Daily* [解放军报], February 3, 2021.

[m] Lü Weijie [吕惟劼] and Feng Guangming [冯光明], "How a 'Steel Tailor' Is Trained" ["钢铁裁缝" 这样练成"], *PLA Daily* [解放军报], November 1, 2024.

[n] Wang Ke'an [王柯鳗] and Li Changhuan [李昌寰], "Helping Them Mount the Horses and Sending Them on Their Way" ["扶上马还要送一程"], *PLA Daily* [解放军报], February 23, 2018.

[o] Chen Guoquan [陈国全], Duan Jiangshan [段江山], and Zhang Miao [张淼], "The Steel Backbone of the Ocean Depths" ["大洋深处的钢铁脊梁"], *PLA Daily* [解放军报], June 20, 2018.

[p] This crew appears to operate a nuclear submarine in the Northern TCN. Sui Yifan [隋一范] and Zhang Dongpan [张东盼], "Strengthening Grassroots Bastions, Setting an Example as the Vanguard" ["强基层堡垒 当先锋模范"], *PLA Daily* [解放军报], June 30, 2024. On the location of this unit, see Zhang Chen [张忱], "Underwater Pioneers Who Fulfilled the Mission with Their Lives" ["用生命践行使命的水下先锋"], *CPC News* [中国共产党新闻], October 28, 2013.

[q] Similar examples can be found in a 2018 article in which the organic maintainer often leverages telephone calls with the military enterprise technician [军工企业技师] while the ship is underway, such as to repair a steering gear system. This technician is also described as having been at the shipyard during the acceptance of the ship [船厂接舰期间] (Duan, 2018).

[r] Xu Qiwei [徐启薇] and Gao Yuan [高元], "A Look Back at the Heartwarming Stories of 2024" ["盘点2024温暖回响"], *PLA Daily* [解放军报], December 28, 2024.

[s] Yang Yunxiang [杨云翔], "Develop a 'Keen Eye' Through Diligent Study and Critical Thinking" ["勤学善思练就"火眼金睛"], *PLA Daily* [解放军报], November 29, 2024.

[t] Fan, Li, and Zhang, 2021.

[u] Li Xiwen [李习文], Li Wenxuan [李汶炫], and Zhang Lin [张琳], "Talented Personnel Emerge in Large Numbers and Vitality Bursts Forth" ["英才辈出活力迸发"], *PLA Daily* [解放军报], July 1, 2021.

[v] Lei Ting [雷霆] and Ni Haoyang [倪浩洋], "Integration into the Teaching Process to Mutually Inspire and Interact" ["融入施教过程 相互启发互动"], *PLA Daily* [解放军报], June 23, 2024.

CHAPTER 2

Factors Shaping Modern Naval Maintenance Management

To aid readers in understanding considerations that inform a modern navy's maintenance management, we begin this chapter with a brief overview of dynamics linked to force structure, infrastructure, and workforce that shape naval maintenance activities. We then examine the relationship between force design, shipbuilding, and maintenance in three navies—the contemporary USN, the contemporary Japan MSDF, and the Soviet Navy as it was during the height of the Cold War. We apply insights from these maintenance archetypes to understand trends applicable to the PLAN's rapid modernization. Next, we review shipbuilding dynamics within China that affect PLAN maintenance management. This chapter sets the scene for the detailed analysis of PLAN maintenance practices that follows in Chapter 3.

Approaches to Naval Maintenance Archetypes

Similar to many aspects of the logistics warfighting function, maintenance can, at its heart, be understood as an issue of supply and demand. Demand for maintenance services is influenced primarily by (1) a fleet's force structure and (2) platform- and component-specific needs. A fleet's force structure refers to the quantity and mix of platforms (aircraft carriers, submarines, surface combatants, etc.); these vessels are all built with an expected service life and will require maintenance and modernization to continue operating for the duration of their service life.[1] The force structure of a fleet is itself the product of several factors—primarily the need to meet the strategic objectives assigned to the naval service and the service's force modernization and procurement strategy.[2] Each vessel and its major components (communications, major combat systems, damage control system, etc.) also have specific maintenance requirements, which are tied to ship and component designs, the ship's operating environment, and the ship's operational tempo.

The supply of naval maintenance similarly has two major components: (1) labor and (2) facilities and equipment. Naval maintenance has historically been a manpower-intensive industry process, many aspects of which require specialized skills, especially to maintain the sensors, communications, and combat systems of modern military ships. Critical factors here are the quantity of personnel (including uniformed navy, con-

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1 Bradley Martin, Michael McMahon, Jessie Riposo, James G. Kallimani, Angelena Bohman, Alyssa Ramos, and Abby Schendt, *A Strategic Assessment of the Future of U.S. Navy Ship Maintenance: Challenges and Opportunities*, RAND Corporation, RR-1951-NAVY, 2017; GAO, 2022a; GAO, *Navy Ships: Applying Leading Practices and Transparent Reporting Could Help Reduce Risks Posed by Nearly $1.8 Billion Maintenance Backlog*, GAO-22-105032, May 2022b.

2 Multiple factors and constraints influence fleet size, but the chief driver is the need for a naval service to meet strategic objectives, which are typically outlined in guidance documents. See, for example, Deputy Assistant Secretary of the Navy (Budget), "Highlights of the Department of the Navy FY2025 Budget," Department of the Navy Office of Budget, February 29, 2024, pp. 1-1–1-11.

tractors, or other employees), the quality of their training, and their experience.[3] Facilities and equipment are also essential to determining naval maintenance capacity: Dry docks and specialized equipment are typically needed for major maintenance cycles and modernization, and replacement ship components (spare parts) are needed throughout a vessel's life cycle. Shipyard capacity, equipment availability, and the production and availability of ship components all play a role.[4]

At a high level, how do these supply and demand factors (and their component influencing factors) interact with maintenance? A review of the maintenance practices of 20th- and 21st-century naval forces suggests several models, or *archetypes*, of surface vessel employment, which serve as a useful frame to help analyze the PLAN's maintenance practices and long-term trajectory. We observed that strategic naval objectives shape countries' naval maintenance and shipbuilding approaches; different navies optimize to achieve different specific objectives amid constraints. The recent histories of the USN's contemporary surface fleet, the Soviet Navy's surface fleet during the height of the Cold War, and the Japan MSDF's contemporary submarine fleet provide three case studies to explore these relationships.

The U.S. Navy, 1990–Present

The USN's ongoing struggles to maintain its fleet of surface ships at a sufficient level to support stated U.S. strategic objectives have roots in both supply and demand factors for naval maintenance. Although the challenges faced by the USN have been well documented and Navy leadership has stated that maintenance improvement is a priority, many identified challenges have yet to be addressed, and delays and cost escalation remain the norm.

Researchers have highlighted how, on the demand side, the historically high rate of surface ship deployment—the result of an expansive set of assigned missions for the fleet—has contributed to increased maintenance requirements—and, therefore, lower ship availability.[5] Relatedly, ship procurement and production capacity remain at levels inadequate to compensate for an aging fleet that requires longer and more-frequent maintenance periods to remain operational, demonstrating the compounding effect of force structure on maintenance requirements without changes to strategic objectives.[6] Furthermore, advanced ship designs have become the standard across the force, making each individual ship more maintenance-intensive.[7]

In terms of supply factors, labor shortages persist as a key influencer of maintenance shortfalls. Such labor issues include manning shortages on surface ships that make it difficult to complete routine and emergency maintenance actions while a ship is underway. This lack of action contributes to an inability to meet maintenance demands on time, potentially raising the risk of greater damage in the long run.[8] These labor shortages can be further compounded by inadequate training on how to perform maintenance, resulting in a less capable force that is unable to address maintenance needs in a timely manner.[9] Labor and equipment shortages at shore-based maintenance facilities that conduct more-intensive maintenance also result in delays and

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3 Diana C. Maurer, "Navy Maintenance: Persistent and Substantial Ship and Submarine Maintenance Delays Hinder Efforts to Rebuild Readiness," congressional testimony before the Subcommittees on Seapower and Readiness and Management Support, Committee on Armed Services, U.S. Senate, U.S. Government Accountability Office, GAO-20-257T, December 4, 2019.

4 Martin et al., 2017; Maurer, 2019; and GAO, 2022a.

5 Jonathan G. Panter, *The Illogic of Naval Forward Presence*, thesis, Columbia University, 2024.

6 Congressional Budget Office, *An Analysis of the Navy's 2025 Shipbuilding Plan*, January 2025.

7 Steve Wills, "End the Navy's 30-Year Slide in Capability and Capacity," *Proceedings*, Vol. 149, No. 4, April 2023; Peter Ong and Bradley Martin, "RAND: What Does the U.S. Navy Need in Its Future Combatants?" *Naval News*, April 25, 2024.

8 Diana Stacy, "How a Sailor Shortage Is Crippling Ship Maintenance at Sea," *Navy Times*, September 11, 2024.

9 GAO, *Navy Readiness: Actions Needed to Improve Support for Sailor-Led Maintenance*, GAO-24-106525, September 9, 2024.

deferment of maintenance that contribute to maintenance backlogs and reduce ship availability.[10] Relatedly, private shipyards are contracted for a variety of maintenance tasks on most surface ships,[11] but scholars highlight the lack of dry dock space and aging equipment at public shipyards in the United States as an issue for maintaining the USN's nuclear-powered fleet of submarines and aircraft carriers.[12]

Although the preceding explanation of USN surface fleet maintenance challenges in the post–Cold War era is not fully applicable to the comparatively youthful PLAN fleet,[13] there are several parallels worth considering. On the demand side, similarities can be found in the PLAN's high operational tempo, including the increasing rate of "far seas" operations, and the growing design and technological sophistication of the PLAN surface fleet over time. On the supply side, comparisons with the USN are possible regarding manning levels and the training quality of PLAN surface ship crews.

The Soviet Navy, Early 1960s–Early 1990s

In the early 1960s, the Soviet Navy rapidly constructed a large number of surface combatants, which began to be employed in large numbers in blue-water exercises by the early 1970s.[14] These ships were simple and cost-efficient compared with USN vessels of the era and were optimized for short-notice, near-shore deployment focused on sea-denial missions. The high rate of production and increase in deployment led to high demand for maintenance.[15]

On the supply side, however, Soviet naval maintenance culture and practices were poor, and the system was heavily reliant on conscripted sailors with short terms, which limited the development of an experienced corps of maintainers. Soviet ship and system designs sought to meet this reality by focusing on reliable, simple, and standardized designs. This suggests that one approach to compensating for maintainer proficiency is to make specific ship design choices, but, in the case of the Soviet Navy, it proved insufficient. A lack of adequate maintenance limited the capacity for the Soviet Navy to compete with the USN throughout the 1980s, particularly as Soviet leaders attempted to maintain a persistent presence in contested areas. By the end of that decade, the Soviet Navy faced extensive readiness challenges that effectively shrank the size of its fleet.

This boom-and-bust modernization arc provides one example of what a trajectory might look like for a surface fleet that rapidly builds ships and employs them at a relatively high tempo in blue-water operations but is limited by maintainer proficiency. How the PLAN develops and trains its maintainer workforce and incorporates and prioritizes maintenance activities in operating cycles will have important implications for the health and longevity of the PLAN surface fleet, especially if it continues to be used at a high tempo in the Western Pacific and beyond.

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[10] GAO, 2022a. The "Homeport Rule" (10 USC 8680) restricts maintenance of U.S. homeported ships to shipyards within the United States or Guam, the intent being to maintain ships near their homeport. Although this is a legislative constraint that prohibits the outsourcing of maintenance to foreign ports, it appears there are enough granted exceptions to the rule that it is not a significant impediment causing maintenance backlogs. However, given the growing maintenance burden for U.S. naval ships and issues with the U.S. industrial base meeting that demand, this rule could have more impact in the future.

[11] See Tim Colton and LaVar Huntzinger, *A Brief History of Shipbuilding in Recent Times*, CNA, September 2002, for a still-relevant assessment identifying issues with commercial shipbuilding that affect naval shipbuilding.

[12] Maiya Clark, "U.S. Navy Shipyards Desperately Need Revitalization and a Rethink," Heritage Foundation, Backgrounder No. 3511, July 29, 2020.

[13] The PLAN does not have the same challenges as the USN in its ship procurement and production capacity; the PLAN also does not appear to suffer from a lack of dry dock space.

[14] John T. Kuehn, "Zumwalt, Holloway, and the Soviet Navy Threat: Leadership in a Time of Strategic, Social, and Cultural Change," *Journal of Advanced Military Studies*, Vol. 13, No. 2, Fall 2022. For more resources on this period, see Central Intelligence Agency, "CIA Analysis of the Soviet Navy," webpage, undated.

[15] Steven E. Miller, "Assessing the Soviet Navy," *Naval War College Review*, Vol. 32, No. 5, September–October 1979.

Japan Maritime Self-Defense Force, Early 2000s–Early 2020s

The Japan MSDF and Tokyo's submarine industrial base provide an alternative archetype with a different approach to balancing supply and demand.[16] The MSDF attempts to balance "cost, readiness, and wartime surge capacity"[17] while seeking to retain a stable workforce. Japan's approach is to rapidly replace and demote aging vessels instead of maintaining older ships in the inventory; emphasis is placed on replacement rather than on investment in mid-life overhauls. Japan's traditional shipbuilding strength allows it to maintain a highly proficient and professional naval workforce. Features of this system include

- stabilizing production systems for new boats (one submarine produced per year)
- decommissioning vessels well before the end of their service life (typically 18 years into a service life of 35–40 years)
- transferring decommissioned vessels to training and reserve fleets, where they are maintained.

These factors allow the MSDF to maintain a stable submarine production industry while meeting the fleet requirements of Japan's maritime strategy. The MSDF could also opt to delay decommissioning of submarines as new production is fielded, thus increasing the size of the fleet. However, Japan's approach likely incurs greater financial costs in peacetime because mid-life overhauls are less costly than new construction. Potential points of comparison with the PLAN are regularized production of large surface combatants and factors related to the shipbuilding industrial base informing naval shipbuilding decisions.

A summary of the attributes for these different archetypes is presented in Table 2.1.

Shipbuilding Dynamics Shaping PLAN Maintenance

China's shipbuilding has rapidly grown in scale and sophistication since the mid-2000s, and that growth that has been accompanied by the development of multiple shipyards and an increased demand for a highly technical and skilled workforce.

Much attention has been given to the links between the PRC's naval and commercial shipbuilding. Commercial shipbuilding rapidly increased prior to the 2008 financial crisis, then fell from 2008 to 2014; a large

TABLE 2.1
Naval Shipbuilding and Maintenance Archetypes

Navy (Period)	Arc of Force Modernization	Repair Investment Strategy	Maintenance Force Characteristics
USN surface fleet (1990–present)	Shrinking—affected by deferred maintenance and service life extensions	Repair	Skilled, but issues with manning levels
Soviet Navy surface fleet (1960–1980)	Boom then bust—emphasis on simple, cost-effective platforms	Build new	Conscript-based; poor maintenance culture
Japan MSDF submarines (2000–2020)	Steady—keeping the industrial base strong, but at higher cost	Replace	Skilled, with emphasis on a stable workforce

[16] We opted to examine the submarine shipbuilding sector as opposed to the surface ship sector because of Japan's apparent emphasis on producing submarines at regular intervals, which is a useful point of comparison with PLAN surface ship production. A review of production data on the MSDF's major surface combatants did not find the same trend of surface ships of a particular type being produced at regular intervals. Potential explanations could include producers shifting between different ship classes or taking up the slack in naval orders with commercial ship production.

[17] Jeong Soo "Gary" Kim, "Japan's Submarine Industrial Base and Infrastructure—Unique and Stable," Center for International Maritime Security, July 15, 2024.

number of naval shipbuilding projects over the latter period compensated for the reduction in commercial demand.[18] Beginning in the mid-2010s, however, the PRC's commercial and naval shipbuilding sectors have become more bifurcated as a result of key trends in each area. The commercial sector has matured and stabilized. Naval construction has begun to shift from large numbers of relatively simple classes of ships, often constructed in a short time, toward relatively small numbers of more-advanced ships, which take longer to build.[19] This shift to producing more-advanced ships has been accompanied by a concentration of constructing specific ship types in only a few shipyards. For example, such ship classes as the Luyang III Type 052D destroyer and Renhai Type 055 cruiser are each built in two shipyards. This trend might reflect the degree of specialization, in terms of both personnel and equipment, required to construct such vessels. It also implies that the PLAN now has higher demands for specialized technical expertise within the maintenance force.

For the PLAN fleet, a ship's service life depends on a variety of factors, such as the shipbuilding industry's technical capability, the ship's structure, and the operational profile. In the 1990s, PLAN ships often exceeded their planned service life because of a lack of funding for new ship construction; since 2000, there has been a reduction in ships exceeding their service life.[20] However, the PLAN continues to study other countries' approaches to equipment life extensions and how they might apply to the PLAN, asking such questions as "How can we scientifically upgrade and transform old equipment to better extend its combat effectiveness [using] limited military expenditures?"[21] For certain classes of ships, there is evidence that the PLAN is investing in mid-life overhauls to restore ships to their initial condition and performance. For example, the Type 052C DDG is scheduled for a service life of 30 to 35 years. At the 17-year mark, members of that ship class have entered into comprehensive scheduled maintenance consistent with a mid-life overhaul. This involves some modernization of the platform, but those efforts are minimal and considered cost trade-offs.[22] The PLAN has also demonstrated a willingness to maintain and overhaul older classes of ships to maintain their operational status, such as the *Sovremenny*-class destroyer, even as more-advanced vessels have been fielded. In recent deployments and exercises, the PLAN has often employed its more-modern and more-advanced destroyers and cruisers rather than legacy platforms.[23]

A third dynamic is that the rapid build rate within the PLAN suggests a significantly higher maintenance demand beginning in the 2030 time frame if the PLAN opts to conduct mid-life overhauls for its newest large surface combatants. Even if the PLAN decides not to conduct overhauls for all advanced surface combat-

18 Predd, Kim, and Carroll, 2024.

19 Predd, Kim, and Carroll, 2024.

20 Ju Lang, 2024.

21 Wang Yue [王越], Fan Weijie [樊伟杰], and Wang Ying [王英], "Life Extension, Giving End-of-Life Equipment a 'New Lease' on Life" ["延寿，让达龄装备重获‘新生’"], *PLA Daily* [解放军报], March 15, 2024.

22 For example, there have been certain efforts as part of the mid-life overhaul to modernize the 052C to 052D specifications for some easily installable shipboard systems. But the PLAN has not upgraded the VLS because doing so would require significant hull cuts that would be as costly as a new ship. See Ju Lang, 2024.

23 John Dotson and Jonathan Harman, "The PLA's 'Strait Thunder-2025A' Exercise Presents Further Efforts to Isolate Taiwan," *Global Taiwan Brief*, Vol. 8, No. 8, April 16, 2025; Dzirhan Mahadzir, "Chinese Navy Task Group Operating in Australia's EEZ; U.S., Japan Hold Ballistic Missile Exercise," *U.S. Naval Institute News*, February 25, 2025; Ian Ellis [@ianellisjones], "JOINT SWORD 2024B BRIEF + MAP," post on the X platform, October 15, 2024; Dzirhan Mahadzir, "Chinese Aircraft Carrier Operating Near the Philippines," *U.S. Naval Institute News*, October 2, 2024; Blasko, 2024; John Dotson and Jonathan Harman, "The PLA's Inauguration Gift to President Lai: The Joint Sword 2024A Exercise," *Global Taiwan Brief*, Vol. 9, No. 12, June 12, 2024; Bonny Lin, Brian Hart, Samantha Lu, Hannah Price, and Matthew Slade, "Tracking China's April 2023 Military Exercises Around Taiwan," China Power Project, Center for Strategic and International Studies, updated November 8, 2023; Eric Chan, "Operationalizing Symbolic Encirclement: A Comparison of PLA Exercises Following Recent High-Profile Visits," *Global Taiwan Brief*, Vol. 8, No. 8, April 19, 2023; Akhil Kadidal and Ridzwan Rahmat, "Chinese Wargames End with Massed Incursions, Carrier Aircraft in Taiwan's ADIZ," *Janes Defence Weekly*, April 12, 2023.

ants, other maintenance activities, driven by regularized maintenance demands and/or specific operational tempo demands, will ask more of the PLAN's maintenance workforce and infrastructure as the PLAN's force structure grows.

To take the PLAN's newest DDGs as an example, the first Type 052C DDG reached the 17-year mid-life overhaul window in fall 2022, and the sixth and last Type 052C DDG would be slated to receive an overhaul beginning in early 2030.[24] This implies that the potential overhaul demand for Type 052Cs is six ships over roughly 8.5 years. In comparison, at least 34 Type 052Ds were commissioned between spring 2014 and the end of 2025.[25] A similar 17-year overhaul time frame implies an overhaul demand for 34 ships over only about 11.5 years, beginning in 2031 and lasting through at least 2042. Adding in the overhaul demand for the Type 055 CG only worsens this picture: Using the same assumptions, we arrive at nine Type 055s commissioned since early 2020 and requiring overhauls between 2037 and 2042.[26] Moreover, construction of both DDG and CG classes continues, meaning that this higher demand for major maintenance activity will continue beyond 2042. These notional schedules imply that maintenance yards will need to add capacity to meet a surge that is solely a function of when the ships were commissioned. After the surge, there could also be a decline in demand that could be difficult to manage. These dynamics could lead to delays in moving ships both in and out of overhaul, and some ships could end up delayed and possibly overused if the ships that are supposed to relieve them end up in the yards for longer than expected.

Conclusion

The interplay of maintenance demand and supply linked to fleet force modernization suggests certain archetypes based on 20th- and 21st-century U.S., Japanese, and Soviet naval forces. For the modern USN, maintenance shortfalls are an ongoing but not insurmountable challenge, whereas the Soviet surface fleet from the 1960s to 1980s experienced a "boom then bust," fielding many new ships that it was ultimately unable to maintain. The PLAN has conducted overhauls for its now-legacy surface combatants and appears to be considering doing so for its newer ship classes, suggesting that is more likely to take a hybrid approach. As Chapter 3 further explores, the PLAN is investing in the skill of its maintenance forces, which also suggests a repair-focused strategy. However, the bow wave of PLAN maintenance probably has yet to crest, given that overhauls for its new destroyer and cruiser classes will likely not begin until in the early 2030s.

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24 Ship commissioning dates in this paragraph are from Janes. See the appendix for more details.

25 Quantities are based on information from Janes updated as of January 26, 2026. In our Luyang III count, we include one ship for which the commissioning date is not available, but the ships launched before and after it have both already commissioned. We exclude from this count another Luyang III DDG that is listed as commissioning sometime in 2026. It will be the 35th commissioned since 2014.

26 We exclude from this count one Renhai CG that is listed as commissioning sometime in 2026. It will be the tenth commissioned since 2014.

CHAPTER 3

Insights on PLAN Maintenance Activities and Management

In previous research on this subject, we devised a framework for assessing organizational attributes and maintenance best practices to characterize and inform trends in other militaries' maintenance practices.[1] For this research, we leverage insights from that framework but venture beyond it to focus on themes related to the growing demand for maintenance, specific demands on workforce expertise, self-sufficiency, and other topics. Overall, we find that the PLAN is investing in improving organic maintainers' capabilities via academic and other partnerships and mentoring initiatives. The PLAN is particularly focused on improving self-sufficiency, meaning the ability to repair and maintain equipment and systems independently of direct reach-back or broader support from external organizations. However, organic maintenance forces continue to depend on support from higher maintenance echelons, such as enterprise technicians, to conduct more-sophisticated repairs.

Sources on PLAN Maintenance Activities and Management

Analysis in this chapter primarily draws on Chinese-language articles published between 2018 and 2024 in the *PLA Daily* and republished on China's defense ministry website; we selected and reviewed more than 90 *PLA Daily* articles focused on PLAN maintenance and logistics activities. To supplement these articles, we leveraged journal articles by PLAN authors in PLA-affiliated journals, such as *World Navy Research* [世界海军研究], *Military Systems Engineering* [军事系统工程], and *Logistics* [后勤]; professional military education texts; and technical and reference books on PLA naval equipment and naval maintenance terms. We also reviewed media articles on broader PLA maintenance activities and relevant military exercises, typically from select Chinese state-media websites and non-authoritative sources, such as the *South China Morning Post*, and English-language secondary sources that have assessed PLA logistics and maintenance practices. In particular, the U.S. Naval War College's China Maritime Studies Institute has published several foundational resources on shipbuilding, naval operations, and maintenance-related topics.[2]

Key Insights

We found three major themes in PLA discussions of PLAN maintenance: the growing demand for maintenance resulting from building more and more-advanced ships, the implications of increasing technology complexity for PLAN maintainer workforce expertise, and efforts to improve self-sufficiency of maintenance activities conducted by ship crews. We examine these and related topics in the following sections.

1 Fleming et al., 2023, pp. 39–56.

2 These include Andrew S. Erickson, ed., *Study No. 6, Chinese Naval Shipbuilding: An Ambitious and Uncertain Course*, China Maritime Studies Institute, 2016; Dutton and Martinson, 2017; Ju Lang, 2024; and many others.

Demand for Maintenance and Repair Already Increasing Because of Rapid Shipbuilding Growth and Modernization

PLA sources clearly recognize that the recent growth in the size of the PLAN fleet will lead to increased demand for maintenance across the life cycles of the PLAN's new assets. For example, the venerable Dalian Shipbuilding Industry Company (established in 1898) produces both commercial and naval ships, including aircraft carriers and the PLAN's most-modern surface combatants—the Renhai (Type 055) CGs and Luyang III (Type 052D) DDGs.[3] Ship orders at Dalian have increased since the 1990s, but the shipbuilding workforce has not grown. The chief quality engineer of the shipyard noted that "in the 1990s, we only built two or three ships a year, but now we can build 20 to 30 ships a year. However, the number of workers has not increased, mainly because of technological progress and the overall improvement in the quality of the workers."[4] Factory workforce dynamics could shape PLAN ship maintenance, given the relationship between PLAN maintenance units and factories, in which maintainers consult factory personnel and also provide them feedback. This feedback process and provision of subject-matter expertise would likely be constrained by the increase in production without a commensurate increase in personnel. Other initiatives, such as efforts to increase PLAN maintenance unit proficiency and strengthen partnerships with universities, could alleviate some of this strain. These relationships are discussed in greater detail in the section titled "Partnerships Between Maintainers, Educational Institutions, and Manufacturers Demonstrate Positive Feedback Mechanisms."

Cost Is Considered as a Factor in Maintenance Decisionmaking

PLA research articles focus on various aspects of ship maintenance from a cost perspective, such as diesel engine repairs—an issue commonly reported on in the *PLA Daily* articles we surveyed. A 2023 article on diesel engine cylinder head repairs notes that challenges include "high demands on the timeliness of maintenance support, long procurement cycles for critical spare parts, low average consumption rate of components, high costs associated with excess inventory of spare parts . . . the single-use nature of worn parts that cannot [benefit] from life extension, and difficulty in recycling worn-out parts."[5] To improve diesel engine maintenance, the authors recommend improving the timeliness of spare parts support, reducing inventory costs for spare parts, and extending the life span of worn parts. They also propose that additive remanufacturing to repair damaged cylinder heads could help reduce maintenance costs.

As systems approach mid-life, the PLAN is also thinking through maintenance and broader sustainment implications. An article on the overhaul of a Type 052C destroyer highlights the bill coming due for ship classes commissioned since the 2000s:

> In the past 20 years, the development speed of China's naval surface ships has attracted worldwide attention. This is a very amazing speed, but it is also a worrying speed . . . [that] is unsustainable. In other words, such a [construction] speed cannot be sustained. . . . While reasonably arranging the shipbuilding plan,

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3 See the appendix for more details.

4 Fu Malin [符马林], Liu Jianwei [刘建伟], and Kang Zichen [康子湛], "The Modern Pulse of a One Hundred-Year Old Shipyard" ["一个百年船厂的现代化脉动"], *PLA Daily* [解放军报], December 14, 2018. The quoted engineer appears to manage both commercial and naval shipbuilding; he is described as leading a team that retrofitted China's first aircraft carrier. Of note, Dalian has typically focused its commercial shipbuilding in a newer shipyard within the broader complex and uses an older shipyard for military construction (Gabriel Collins and Michael C. Grubb, *A Comprehensive Review of China's Dynamic Shipbuilding Industry*, China Maritime Studies Institute, August 2008, pp. 25–26). Our review of satellite imagery from 2019 to 2025 suggests that large surface combatant construction occurred at the same facility within Dalian over that period. See the appendix for more details on Dalian's naval ship production data.

5 Wen Xuedong [文学栋], Wang Weiqing [王为清], and Dong Enqing [董恩庆], "Research on Diesel Engine Cylinder Head Repair Based on Metal Additive Remanufacturing Technology" ["基于金属增材再制造技术的柴油机气缸盖修复技术研究"], *World Navy Research* [世界海军研究], No. 2, 2023, p. 49.

> we should also pay attention to the extension of the service life of the active ships and the "freshness" of the combat technology standards. Separating the construction, overhaul, upgrade, and decommissioning cycles in an orderly manner is conducive to budget stability, which is an important part of sustainable development. . . . After all, the PLAN needs to compete with the most powerful sea and air forces on earth. How to use limited resources as efficiently as possible to improve competitiveness is the key.[6]

The article demonstrates that the PLAN is carefully considering trade-offs related to service life extensions after observing what the USN and other modern militaries are doing to sustain their fleets. Service life is considered in PLAN shipbuilding. From 1990 to 2000, ships often exceeded their life span because of funding constraints that limited new ship construction. Since 2000, however, the number of ships operating beyond their planned service life has been reduced.[7] The PLAN recognizes that service life will depend on the proficiency of the shipbuilding industry, ships' structural strength, and wear and tear. These factors directly relate to what will be required for repairs and maintenance later in the ships' life cycles. The high rate of production since the mid-2000s means that a larger repair and modernization bill will come due in the future, as discussed in Chapter 2—and those bills could be especially large if the PLAN further considers the need for ship life extensions in the future.

The PLAN is also making efforts to adapt lessons learned from other militaries by planning for service life extensions and modernization. For example, one article describes how the PLAN has been studying recent U.S., United Kingdom, Canadian, Russian, and German decisions to extend the service life of aging vessels and other assets, assessing that the key trade-off is improving combat effectiveness while managing cost.[8] The article notes that these decisions affect maintenance because "the maintenance frequency of equipment after life extension is [generally] higher than that of equipment in regular service, and the number of maintenance items is greater than that of equipment in regular service."[9] There is also interest in floating docks—their mobility could allow them to provide more flexibility as maintenance and repair demands evolve.[10]

Shipbuilding Growth Places Stress on Other Maintenance-Related Aspects

The growing size of the PLAN fleet is putting pressure on maintenance capabilities for other PLAN assets in addition to surface ships, such as aircraft carriers and carrier-based aviation. In one PLAN Aviation regiment's repair shop [海军某团修理厂], "the number and type of aircraft in service with this regiment have continued to increase . . . in recent years . . . and the demand for professional maintenance personnel has become more urgent by the day."[11] The previous training cycle for new junior maintainers had taken a year or more, but the repair facility sought to reduce that length as a way to meet the increased demands of maintaining the regiment's aircraft. The facility shortened the cycle to only six months by implementing a "mentor responsibility system" in which each junior maintainer is paired with a senior colleague who supplements class-based learning with scenario-based exercises and question-and-answer sessions. The pairs also attend monthly maintenance innovation seminars together.[12]

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[6] Ju Lang, 2024, pp. 22–24. The original Chinese-language article dates from 2022.

[7] Ju Lang, 2024.

[8] Wang, Fan, and Wang, 2024.

[9] Wang, Fan, and Wang, 2024.

[10] Guo Wencong [郭文聪] and Yan Ziyi [晏子祎], "The Ship's General 'Hospital'" ["舰艇的全科 "医院""], *PLA Daily* [解放军报], November 29, 2024.

[11] Wang Huangfei [汪黄飞], "The Growth of New Maintenance Troops Has Started to 'Accelerate'" ["机务新兵成长开启 "加速度""], *PLA Daily* [解放军报], October 22, 2024.

[12] Wang, 2024.

The high pace of shipbuilding has led to challenges in scheduling vessel training as ships enter the force after they are built or reenter the force after overhauls. VTCs are specialized organizations that provide basic training for new or recently repaired vessels to ensure their deployability. A vessel's time at a VTC has important implications for maintenance, because once a ship passes its evaluation, it either gains (for new ships) or regains (for recently repaired ships) its Class I deployable status and is sometimes immediately deployed.[13] To keep pace with the surge in shipbuilding, VTCs have adopted a rolling admissions policy rather than admitting vessels once per year. For example, from 2002 to 2007, the Eastern VTC increased the number of ships it trained per year from two to 16.[14] To accommodate this growth, the training cadre combines some ship crews to be more efficient, even though the focus is on one ship. The number of training staff, rather than the number of training facilities, is often the limiting factor to increasing the number of ships trained. The rapid pace of new ships being built has taxed trainers as the technology has become much more sophisticated.[15] Often, instructors' knowledge is "quickly rendered obsolete" by the fast pace of fleet modernization.[16]

Along with experiencing force-level growth, PLAN assets are increasingly operating farther from China's shores. An article on ship-based electromechanical engineers linked the force structure increase and the PLAN's expanding area of operations to higher expectations for the expertise of ship-based maintainers: "[W]ith the commissioning of new warships, our country's navy continues to move toward the 'deep blue'" of far seas operations. A senior marine engineer stated that "a navy heading for the deep blue must have electromechanical sailors who can adapt to the deep blue."[17] Changes in operational tempo and operational reach will affect the maintenance requirements for the PLAN fleet.

Despite Emphasis on Repair, Full Capacity to Implement Might Be Lacking

An emerging theme in writings about PLAN maintenance is an emphasis on repairing items rather than replacing them, even for small issues. In one extreme example at a naval observation and communication station in the Eastern TCN, veterans of the station repair the small capacitors in light bulbs rather than replace them. As the article states, in this unit, "if something breaks, everyone's first thought is to repair it rather than throw it away."[18] It is unclear how widespread this practice is—there is potential that, because of the remote location of the unit,[19] there is extra effort to repair even the smallest component—but the article does indicate an attempt to praise this practice to encourage a shift in thinking.

In an example more similar to efforts being pursued by the USN, an avionics technician in a regiment of the Naval Aviation University repairs helicopter circuit boards. He observes that he should "dissect first, then

[13] Martinson, 2022.

[14] Martinson, 2022.

[15] Increasing pressure appears not to be limited to just maintenance personnel and infrastructure; broader support units and related infrastructure are feeling the crunch as well. A 2021 article quotes the political commissar of a Southern TCN service support *dadui*/squadron as comparing the rapid commissioning of new ships with dumplings: "As more and more 'dumplings' [ships] are being launched, the 'digestion rate' of naval ports has become a new challenge." The commissar added that the demand for logistical support in terms of food, fuel, ammunition, and materiel support had correspondingly all increased significantly, and that the traditional support model was not able to meet these new demands. Fan, Li, and Zhang, 2021.

[16] Martinson, 2022.

[17] Jiang, Wang, and Jia, 2024.

[18] Zhang Rongrong [张容瑢], "There Is a Ray of Light Shining in My Heart" ["有一道光芒照映心间"], *PLA Daily* [解放军报], April 1, 2024.

[19] PLAN observation and communication stations [观通站] might not benefit from some of the improvements that have supported surface fleet modernization and support since they are often located on small coastal islands or remote mountaintops. These facilities also have a relatively small footprint, implying limited storage for equipment and supplies. Thanks to Josh Arostegui for raising this point.

repair" instead of "directly replace," partly to save on maintenance costs.[20] The avionics technician conducts these types of repairs by hand, comparing the parts with diagrams. In the USN, the Miniature/Micro-Miniature Module Test and Repair program provides workstations and diagnostic tools to assist sailors in repairing circuit card assemblies. These capabilities allow repairs to happen on ship and without need of external technicians or depot-level repair, increasing readiness and saving the USN money.[21] It is also an attempt to reverse the USN's broader culture of prioritizing replacement over repair. Judging from the articles mentioned earlier, we discern that the PLAN might be pursuing similar efforts, but it is unclear whether the PLAN is using diagnostic tools and workstations to aid these efforts.

Additional Demands on Maintenance Personnel Stem from Increasing Technological Complexity Aboard Ships

PLA sources indicate that the growing technological complexity of ships—including from a design standpoint but also with respect to subsystems—has implications for maintenance and repair. Multiple articles reference new or upgraded versions of systems that increase the complexity of maintenance and repair tasks, including radars and other subsystems. One profile on a hero radar maintainer[22] states,

> In recent years, the ship radar field has developed rapidly, and intelligence and integration have greatly increased the difficulty of equipment maintenance. Liu Xiaojie [the radar technician] said that in order to truly keep pace with equipment development and ensure that radar can "see farther and . . . more clearly," it is necessary to unceasingly acquire new knowledge.[23]

Another radar technician working on PLANMC ground vehicles had more than ten years of experience and previously won first place in a maintenance skills competition, but when the brigade that his unit supports upgraded to new radar equipment, "the components and operational employment of the new radar [were] completely different from the previous radar. He had to become a 'primary school student' again and embark on a new journey of seeking knowledge."[24] The same article chastised other technicians for wanting to return the new radar to the factory [返厂] for maintenance while praising the featured technician for instead solving the issue locally, partly by studying the new equipment "day and night."[25]

As another senior maintainer with nearly 30 years of experience explained to junior technicians, major subsystem upgrades for surface combatants result in maintainers needing to completely refamiliarize themselves with the layout of key systems so they can perform their work. The maintainer, serving aboard the Type 052C destroyer *Haikou*, explained, "every time the ship is repaired or modified, the pipeline path is not necessarily completely consistent with the original drawing. I will always recheck and redraw the pipeline

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20 Zhang Tianqi [张天琦] and Luo Runze [罗润泽], "Aircraft Maintenance Anatomy Expert" ["机务"解剖专家""], *PLA Daily* [解放军报], April 19, 2024.

21 Naval Sea Systems Command, "2MMTR Program Enhances Navy Readiness Through Self-Sufficiency, Cost Savings," April 10, 2024.

22 For more information on *hero maintainers*, see Fleming et al., 2023.

23 Yang, 2024.

24 This anecdote might not fully represent PLAN capabilities; PLANMC technicians typically attend the same schools as PLAA maintainers. Thanks to Josh Arostegui for raising this point. However, the increasing challenge of maintaining more–technologically sophisticated equipment reflects a common theme we have now seen across PLA ground, naval, and air forces. Zhang Feilong [张飞龙], Xia Zhenhua [夏泽华], Mo Haiming [莫海明], and Teng Chunfeng [滕春峰], "The 'Craftsman Spirit' of Combat Vehicle Repair Troops" ["战车修理兵的'工匠精神'"], *PLA Daily* [解放军报], November 19, 2024. This incident is described as taking place "a few years" after the technician won a 2016 competition.

25 Zhang, Xia, et al., 2024.

path to figure out the actual condition of the warship."[26] Another article we reviewed about system upgrades affecting maintainers mentions "many upgrades" between 2009 and 2018 to command and control systems on the same ship (*Haikou*);[27] we also found mentions of upgrades to more-sophisticated electronic components.[28] It is difficult to assess how widespread this issue is across the fleet. Articles noting the maintenance burden of these upgrades suggest that some PLAN crews address them at the individual ship level.[29]

As new ship classes enter the force, added complexity at an aggregate level could further shape maintenance demand. An article on the Dalian Shipbuilding Industry Company describes a team of engineers reviewing China's first aircraft carrier, the *Liaoning*, which was imported from Ukraine. The personnel examined the vessel "inch by inch" to develop "tens of thousands of test plan diagrams" to support the retrofit.[30] More-complex ships also require significant periods for outfitting prior to commissioning. An article on naval surface combatant construction states that building a "medium-sized," "ski-jump" aircraft carrier—China's indigenously produced *Shandong* aircraft carrier is likely a good example—takes more than two years; an additional year or more might be needed for outfitting after the ship is launched. The 10,000-ton class ships, such as the Type 055 cruiser, which is China's most advanced large surface combatant, need up to two years for outfitting. One way China and other navies shorten these timelines—and try to reduce costs—is to conduct "pre-outfitting" for aircraft carriers prior to launch.[31] But having factory personnel spend more time on preparing ships for commissioning could leave them with less time to engage with PLAN maintenance counterparts.

This dynamic might extend beyond surface ships; maintainers of some submarines have transitioned from working on aging diesels to new classes with air-independent propulsion (AIP). In one anecdote about an AIP submarine, a senior NCO proposed to extend the maintenance cycle of a core component of the AIP system, which was initially viewed as "challenge to authority," given that the component's maintenance period was based on years of testing.[32] The NCO persuaded the other attendees at an overhaul coordination meeting (attended by personnel from an academic institution, a research institution, the manufacturer [厂家], and the submarine) that the change would reduce the submarine's maintenance downtime and therefore improve its at-sea rate.[33] After the changes were approved by the flotilla's leadership, the maintenance cycle of the component was extended more than three times. Although this flotilla (responsible for approximately eight submarines) experienced a marked improvement, it is also noteworthy that the change was described as being only flotilla-wide rather than an improvement that was implemented fleet-wide.[34]

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26 Peng Fan [彭镡], Yang Jie [杨捷], and Qian Xiaohu [钱晓虎], "Stick to the Deep Compartment and Chase the Waves" ["坚守深舱逐浪行"], *PLA Daily* [解放军报], December 27, 2024.

27 Fan Jianghuai [范江怀], Chen Guoquan [陈国全], Duan Jiangshan [段江山], and Wang Dong [王栋], "Haikou Ship: Ship in the Spring of Youth" ["海口舰：青春之舰"], *PLA Daily* [解放军报], July 29, 2018.

28 Duan, 2018.

29 For an example of the impact of new equipment on shipboard maintenance, see Yang, 2024.

30 Fu, Liu, and Kang, 2018.

31 Liu Gang [刘钢] and Liao Fuping [廖富平], "What Is the Warship Busy with These Days?" ["军舰这段时间在忙啥"], *PLA Daily* [解放军报], August 12, 2022.

32 Fan, Liu, and Dai, 2018.

33 A different article about a rudder signal technician aboard a submarine highlights that this concern about at-sea rates might be true for submarines; the article indicates short excursions. The technician delays kidney stone surgery to complete a deployment, and his mother calls him while he is convalescing at the hospital. The article notes that when submariners go out to sea, "it is common to be out of contact [with their families] for ten days or half a month." Xiang Liming [向黎鸣], "Ordinary Things Are Worth Recording" ["平凡值得被记录"], *PLA Daily* [解放军报], November 13, 2024. For other anecdotes about the lengths of submarine excursions, see Kennedy, 2024.

34 We thank Josh Arostegui for raising this point.

Giving the rising demand for maintenance across more-complex vessels, some maintainer crews have innovated to reduce repair timelines. The maintenance unit for one naval *dadui*/squadron was experiencing increased demand for maintenance and support because of "the continuous commissioning of new ships"; when parts became damaged, the unit had to wait for replacements to be delivered to the ship before repairs could be conducted.[35] In one instance, the delivery took nearly a day, almost delaying a ship departing port for a mission. At the "urging" (or more likely, censure) of the unit's Party Committee, the maintenance unit put together a "maintenance treasure box" of more than 20 common maintenance tools, more than 50 commonly used spare parts, and a maintenance manual to help crews quickly address common faults without waiting for parts to be sent from shore.[36]

Looking ahead, some articles indicate interest in leveraging new technologies to help improve maintenance capabilities. One example is digital twin technologies to support predictive maintenance and to test methods of equipment life extension.[37] Artificial intelligence technologies can support life extension decisions and fault detection, diagnosis, damage assessment, and remote maintenance.[38]

Self-Sufficiency Prioritized in Maintenance Approaches but Unrealized

Ship maintenance is a costly endeavor that has significant impact on operational availability for naval assets. It is estimated that in most armed forces, one-third to one-half of critical assets are unavailable at any one time because of maintenance requirements.[39] Maintenance best practices optimize for cost and availability; military organizations are incentivized to ensure efficient and effective maintenance practices. Self-sufficiency—meaning the ability to repair and maintain equipment and systems independently—is prioritized in modern navies because it reduces dependence on external agencies, can reduce cost, and increases resilience. Self-sufficiency of maintenance, particularly when underway, increases a ship's ability to stay on station longer. This is important in a high-end fight or as militaries deploy farther abroad. Self-sufficiency relies on the skill level of the maintainers, achieved through education and training, and a culture that supports and prioritizes it. Although self-sufficiency is prioritized in certain naval constructs, a navy can still benefit from increases in efficiency if it successfully leverages the civilian sector. Therefore, an effective use of both forms of support balances operational timeliness with efficiency across the life cycles of key platforms.

Organic Maintenance Is Dependent on Enterprise and Higher Echelons Providing Reach-Back Support

The PLAN largely divides its maintenance tasks between organic maintainers and enterprise technicians but has a dependency on reaching back to higher maintenance echelons to conduct more-sophisticated repairs.[40]

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[35] Ye Yunhan [叶昀翰] and Fa Jiangcheng [法将程], "On the Birth of the 'Maintenance Treasure Box'" ["'维修百宝箱' 诞生记"], *PLA Daily* [解放军报], September 13, 2024.

[36] Ye and Fa, 2024.

[37] Wu Xianyu [吴宪宇], Xu Wei [许伟], and Li Rui [李芮], "Digital Twin Technology Accelerates the Rewriting of Future Wars" ["数字孪生技术加速改写未来战争"], *PLA Daily* [解放军报], September 27, 2024; Wang, Fan, and Wang, 2024. For a broader discussion on interest in digital twin, Internet of Things, and other technologies, see Wu, Chen, and Li, 2023.

[38] Wang, Fan, and Wang, 2024.

[39] Colin Shaw, "Mastering Military Maintenance," McKinsey on Government, Spring 2010.

[40] We reviewed a limited number of sources discussing PLA civilian personnel [文职人员] performing maintenance duties, including a 2022 article about "the first batch of civilian personnel recruited from outside the military" serving in a PLAN unit. This article mentions an engineer transitioning from a tech company to conducting shipboard repair and support. See Zhang Xinyu [张欣玉] and Yang Yongpeng [杨永蓬], "Please Find Attached a Report on the Growth on the First Cohort of Civilian Personnel Recruited from the General Public by a Certain PLAN Unit" ["海军某部首批社招文职人员的成长报告，请查收"],

The PLA seeks broadly to optimize its equipment maintenance support system between troop support and military industrial enterprise, although it has not yet fully done so. PLAN authors writing about maintenance support note that, organically, PLA maintenance support organizations are responsible for "troubleshooting, testing, minor repairs and medium equipment repairs in peacetime, and associated support tasks in wartime."[41] Industrial enterprises help supplement this capability by performing more-sophisticated fixes on high-tech equipment. However, there is recognition that there is a need to integrate these components of the maintenance apparatus to "make up for the 'demand gap' in terms of time and space of troop support."[42] Critics of the process describe it as a "monopolistic repair model" in which the system's developer singularly provides its maintenance support. This, the authors complain, leads to "a certain degree of resource waste and 'laziness' in maintenance practices."[43] Issues impeding military-industry integration in maintenance include lack of communication and coordination across key stakeholders, lack of oversight and supervision, and insufficient technical exchange between military and civilians.[44] PLAN sources identify the potential for dependency on enterprise technicians to further deepen as the PLAN grows and rapidly commissions numerous ships "like dumplings [one after the other]."[45]

The PLAN recognizes the need for self-sufficiency in its maintenance practices and has identified a lack of organic maintenance capacity as a continuing shortfall. For example, an instructor in the Northern Theater Command summarizes this perspective:

> As managers and users of boat and vessel equipment, we must not only know how to be proficient in using the equipment but also possess basic repair skills. We cannot expect remote support from factory technicians [厂家师傅] when on the battlefield. Therefore, during peacetime, we should use training and exercise tasks as a guide and strive to have a clear understanding of equipment performance and accurate fault detection to ensure that the equipment is always in good condition for rapid deployment and combat readiness.[46]

The theme of transitioning to greater reliance on organic maintenance capability and less reliance on civilian technicians conducting naval maintenance is one that persists throughout much of the PLAN's reporting on itself and appears to be continually evolving. In recent years, the PLAN has employed a model whereby contractors and some military technical specialists are positioned at forward operating areas to maintain deployed ships where onboard expertise is insufficient to handle normal repairs. This model

PLA Daily [解放军报], September 9, 2022. It is possible that some of the "technical backbones" [技术骨干] we saw referenced in various articles include these civilian personnel. Either way, the role of these civilians is worth watching going forward. On recent PLAN recruitment of civilian personnel, see "Announcement of Public Recruitment of Civilian Personnel by the PLA Navy Dalian Naval Academy" ["海军大连舰艇学院面向社会公开招考文职人员预告"], March 24, 2020; and People's Liberation Army Navy Political Work Department Civilian Personnel Office [海军政治工作部文职人员局], "2023 Navy Written Exam Waived in Recruitment, Direct Selection of Civilian Personnel" ["2023年海军免笔试招录、直接引进文职人员"], undated. Thanks to Josh Arostegui for raising this point and flagging these sources. For more on civilian personnel in the PLA, see Shanshan Mei and Dennis Blasko, "Back to the Basics: How Many People Are in the People's Liberation Army?" *War on the Rocks*, July 12, 2024.

41 Zhang Bei [张蒂] and Zhu Wei'an [朱伟岸], "Discussion on Promoting the Civil-Military Coordinated Equipment Maintenance and Support" ["深化军地协同装备维修保障建设"], *World Navy Research* [世界海军研究], No. 2, 2023, p. 47. The term *jundi* [军地] in this article title is not the same as the characters used in "military-civil fusion" [军民融合] and could also be translated as "military-local," as in "local authorities."

42 Zhang and Zhu, 2023, p. 47.

43 Zhang and Zhu, 2023, p. 45.

44 Zhang and Zhu, 2023, pp. 45–46.

45 Duan, 2018.

46 Zhou and Xu, 2021.

involves military enterprise technicians on board to consult, including during long-distance escort missions, which is a shift from enterprise technicians historically interacting with organic unit technicians [部队专业骨干] only during the installation stage.[47] The deployment of military technicians on board is criticized as a "nanny-style" approach and a "crutch" that leaves the crew "overly reliant" on the enterprise technicians.[48] More recently, it appears that the PLAN has further integrated the support force system for far seas deployments via a new model that "combines formation-accompanying autonomous support, remote maintenance technical assistance, and emergency forward assistance."[49] This shows a trend toward less dependency on onboard technicians but does not indicate whether organic maintenance has achieved desired levels of self-sufficiency.[50]

Although some gains have been made in organic maintenance capability, many of the repairs referenced are smaller repairs that still give maintainers trouble. One article characterizes these activities as follows: "[D]uring routine training, if there is even a little 'headache' in the equipment, the unit will contact enterprise technical personnel for remote guidance."[51] Some of the repairs cited, such as fixes to steering gear and equipment inspections, seem to be normal troubleshooting rather than complicated repair. The PLAN continues to emphasize that highly technical items are difficult to repair and that capacity-building is slow. Some PLAN officers assess that "maintenance capabilities are slowly improving" through the use of "master-apprentice pairs," such as (in one example) between engineers from two local military equipment repair factories [军队装备修理厂技术骨干] and selected officers and soldiers from a support *dadui*/squadron.[52]

Some articles indicate that units' organic maintainer skills have improved; personnel no longer need to call for help or rely on the accompanying onboard enterprise technicians to complete repairs.[53] Some organic personnel might have been forced to become more self-sufficient because of the coronavirus pandemic. One article relays a 2021 anecdote about the Southern TCN destroyer *Wuhan* suffering a main gun failure during an exercise, forcing it to exit the exercise ahead of schedule, after which a shipboard technician finally solved the problem.[54] The article explains that "due to normalized epidemic prevention and control, manufacturers [厂家] have been even less able to repair onboard equipment in a timely manner, [so] similar incidents to the

47 The term used here is literally "unit expert [or specialized] backbones." Later in the article these personnel are also referred to as "unit technical personnel" [部队技术人员]. Wang and Li, 2018.

48 Duan, 2018. The USN also makes use of technicians in maintaining specific high-tech systems or new combat systems; these are typically civilian technicians. Concerns about overreliance leading to a decline in maintainer proficiency for uniformed sailors is not unique to the PLAN, but our research indicates some greater concern within the PLAN that might not be similar to that in the USN.

49 Zhang Kefeng [张科峰], Yin Liuyu [尹柳玉], and Qian Xiaohu [钱晓虎], "Far Seas and Remote Areas, System Support Is Becoming More Effective" ["远海远域，体系保障更高效"], *PLA Daily* [解放军报], January 3, 2024.

50 On the development and use of networks for remote maintenance technical support, see Wu Fangbing [吴方兵], Chen Dianhong [陈典宏], and Zhou Yancheng [周演成], "A Certain Flotilla of the Southern Theater Command Navy: Full-Time Data Monitoring Is Driving Changes in Equipment Support" ["南部战区海军某支队：全时数据监测催生装备保障之变"], *PLA Daily* [解放军报], February 19, 2022; Wu Dengke [武登科], Xiang Jun [向军], and Yuan Lei [袁磊], "Support Setup and Application of Ship Formation Basing on CRM Management System" ["基于CRM管理系统的舰艇远程技术支援体系与应用"], *Ship Science and Technology* [舰船科学技术], Vol. 38, No 5, 2016; Li Ke [李科], Zhang Jundong [张军冬], and Hu Jinhua [胡金华], "Discussion on Remote Technical Support System of Navy Equipment Maintenance" ["海军装备维修远程技术支持系统体，系架构探讨"], *Ship Electronic Engineering* [舰船电子工程], Vol. 31, No. 8, August 2011. We thank John Chen for suggesting some of these sources.

51 Wang and Li, 2018.

52 Zhang Yi [张毅] and Lü Weijie [吕惟劼], "By 'Carrying the Bags to Train,' Learning New Skills of Support" ["拎包跟训" 学到保障新本领"], *PLA Daily* [解放军报], October 26, 2023.

53 Duan, 2018.

54 Fan, Li, and Zhang, 2021.

Wuhan ship's crew repairing equipment malfunctions by themselves have frequently occurred."[55] However, other articles still describe a requirement for nonorganic support.[56] This might indicate that the PLAN is a certain distance from its idealized self-sufficient state.

Personnel Training and Mentorship Models Among Efforts to Improve Self-Sufficiency

Achieving self-sufficiency requires high levels of maintainer proficiency, something the PLAN is investing in through both informal and formal training, particularly at the NCO level. Within the naval equipment support system, changes have focused on intentional partnerships between senior engineers and junior maintenance personnel to increase knowledge and proficiency. These "two-high teams" employ senior engineers and senior NCOs as team leads to mentor junior maintainers.[57] The stated impact of this approach in a pilot program was an improvement in temporary repair effectiveness, so it is difficult to determine how widely these practices have been adopted across the fleet. The pilot program, as described in one 2023 article, took place at bases that house aircraft carriers, submarines, and destroyers.[58] Other programs are aimed at increasing cross-integration of maintenance practices, as, historically, maintenance has focused on singular skills. To address this, a naval carrier–based aviation unit has implemented a "team leading team" model that emphasizes mutual learning of maintenance skills to increase cross-integration.[59] This model was accompanied by other process revamps, such as development of ship-based support plans, implementation of a maintenance support command and control system, and maintenance crew training and qualification certification. If fully implemented and sustained, these initiatives could constitute serious efforts to improve senior enlisted technical skill and supervisory capability.

Dogmatic Standards Might Detract from Self-Sufficiency Efforts

There appears to be a tension within the PLAN between emphasizing dogmatic adherence to standards and promoting innovative problem-solving. We found evidence that the PLAN seems to be moving away from some of its more dogmatic approaches. For example, there is more emphasis on maintenance training to incorporate frontline experience and interactive methods.[60] Recent policies have focused on reducing memorization requirements and on improving inspections, assessment, and evaluation.[61] PLA-wide guidance since 2019 on talent management changes that are relevant for maintenance has sought to develop a more "fault-tolerant" system that allows junior personnel to make mistakes without fear of consequences.[62] As a

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[55] Fan, Li, and Zhang, 2021.

[56] Lü and Feng, 2024.

[57] Cheng Junmo [程钧谟] and Xie Jia [谢佳], "Promote the Rapid Improvement of Regional Equipment Temporary Repair Support Effectiveness" ["推动区域装备临修保障效能快速提升"], *World Navy Research* [世界海军研究], No. 3, 2023, p. 39.

[58] Cheng and Xie, 2023, p. 40.

[59] Zhang Dongpan [张东盼], Kang Chen [康晨], and Wang Yuanlin [王园林], "From 'Master Leading Apprentice' to 'Team Leading Team'" ["从"师傅带徒弟"到"团队带团队""], *PLA Daily* [解放军报], November 20, 2023.

[60] Chen Zesheng [陈泽生], He Shengfa [何升法], Bao Yong [包勇], and Xiang Liming [向黎鸣], "Interpreting the Torpedo Support Officer's 'Battle Exam Paper'" ["解读鱼雷保障员的 "战位答卷""], *PLA Daily* [解放军报], September 3, 2024; Lei and Ni, 2024.

[61] Cong Rongji [丛镕基], Jiang Xinhui [姜鑫辉], Zhang Shishui [张石水], Shi Tao [石涛], Feng Guangming [冯光明], and Yan Yijin [闫艺瑾], "Helping the Troops Reduce the Burden of Unnecessary Memorization" ["帮官兵减掉不必要的背记负担"], *PLA Daily* [解放军报], October 29, 2024.

[62] Li, Li, and Zhang, 2021.

PLAN squad leader acknowledged, "Past mistakes should not become a burden for moving forward."[63] Use of forums in which sailors share lessons learned from these mistakes have become more prevalent and emphasize sharing knowledge. Historically, personnel might have left the service because they were concerned about the consequences of making a mistake. These new initiatives might be aimed at helping increase retention even when mistakes are made, suggesting the development of a more fault-tolerant system.

However, some articles still emphasize that fixing issues at the grassroots level—that of units at or below the regiment level—will require strict adherence to laws and regulations.[64] Some processes involving both inspection and maintenance activities are overly bureaucratic, such as requiring six different departments to sign off on maintenance for basic fire safety equipment.[65] Superficial inspections increase workload for maintainers and leave the grassroots maintenance personnel frustrated. One article posits that an emphasis on dogmatic adherence to standards that are not relevant to the equipment is causing this problem and that more emphasis should be put on problem-solving.[66]

The PLAN has invested in measures to increase maintenance self-sufficiency, but units are still plagued with surface-level maintenance practices at the grassroots level. A PLAN maintainer studying at an equipment factory [某装备工厂] was accused of "treating the symptoms instead of the root cause,"[67] indicating a shallow understanding of systems and the maintenance they require. PLAN inspections reveal minor faults, particularly for rarely used equipment, such as "cracking," "missing pieces," and parts exceeding the storage period.[68] Low use rates of some PLAN equipment had also resulted in subpar maintenance because of crews' lack of familiarity with equipment performance.[69] Some of these issues are driven by the technical complexity of new weapon systems that require more in-depth repair, which might not be possible given the average level of organic technical expertise. But other factors could involve a risk-averse maintenance culture, as we observed in our research on the PLAA and PLAAF.[70] Organic maintenance expertise not keeping pace with technically complex new classes of equipment is a common issue in many military organizations, including the USN.

Other Insights

This section captures some other trends and dynamics we observed regarding regional maintenance practices, partnerships that support PLAN maintenance practices, and morale issues.

63 Gao Dezheng [高德政], "'Good Sailor Forum': Talk About the People and Matters at Hand" [""好水兵讲坛"：讲好身边人身边事"], *PLA Daily* [解放军报], August 6, 2024.

64 Zhang Xiaohe [张潇赫], Jiang Zhanpeng [蒋展鹏], Tao Fuchun [陶富春], Gao Qun [高群], Yao Chengzhi [姚成志], and Tao Lei [陶磊], "Start with the Details to Implement to the End" ["从细节入手　抓末端落实"], *PLA Daily* [解放军报], January 16, 2024.

65 Wang Shuye [王淑叶], Zhang Zhen [张震], and Niu Tao [牛涛], "One Maintenance Form Affects Six Departments?" ["一张维修表，牵动六个科?"], *PLA Daily* [解放军报], June 5, 2020.

66 Huang Jinyi [黄金艺], Yang Mingyi [杨明宜], Liu Wei [刘伟], He Xiaogen [何小根], Cong Rongji [丛镕基], and Wang Zijie [王子杰], "Make Inspection and Supervision More Practical" ["让检查督导作风更实一些"], *PLA Daily* [解放军报], October 22, 2024.

67 Zhang Yi [张毅], "Strengthen the Foundation of Capabilities and Build Up Confidence in Fulfilling Duties" ["夯实能力基础　蓄足履职底气"], *PLA Daily* [解放军报], August 3, 2024. For a submarine example, see Chen et al., 2024. For an example in which a sailor was lauded as one of the few to perform root cause analysis, see Peng, Yang, and Qian, 2024.

68 The equipment referenced are "gas masks, compasses, and other equipment" and tires. The article also notes that optics and chemical defense equipment are uncommon equipment often overlooked in maintenance. Zhang, Jiang, et al., 2024.

69 Li and Zhang, 2023.

70 Fleming et al., 2023.

Regional Variation Across Theater Navies Shapes Maintenance Practices

Although articles on PLAN maintenance practices do not often state which of the PLAN's three fleets is being discussed, we were nevertheless able to identify some dynamics that could shape maintenance demand differently across the three TCNs. This variation might stem from mission-related factors, geography, and other considerations that affect operational tempo and broader readiness rates. Regarding the mission-related factors, each TCN supports its respective Theater Command's specific strategic orientation. These are listed here in fleet protocol order:

- The **Northern TCN** is subordinate to the Northern Theater Command, which "is responsible for operations along China's northern periphery and border security associated with North Korea[,] Russia, and Mongolia";[71] the U.S. Department of War also notes that Northern TCN "defends China's northern maritime approaches and can also provide mission-critical assets to support . . . naval operations outside . . . [the Theater Command's] area of responsibility in various contingencies."[72]
- The **Eastern TCN** is subordinate to the Eastern Theater Command, which has responsibility for "operations in the East China Sea and against Taiwan."[73]
- The **Southern TCN** is subordinate to the Southern Theater Command, which "is responsible for operations in the South China Sea and support to the Eastern Theater Command during a Taiwan contingency."[74]

How might these differing roles and responsibilities shape maintenance requirements and demands? Here is a brief snapshot as reflected in the PLA articles we reviewed.

For the **Eastern TCN**, as discussed in Chapter 1, many of the PLAN's newer surface combatants have taken place in large-scale joint exercises near or surrounding Taiwan since 2022.[75] We did not find any authoritative guidance on the objectives or requirements tied to fleet readiness for any of the three fleets, but there are indications that some of China's newer ships in the Eastern TCN are quite busy. In an article discussing maintainers and other crew members aboard the *Xiamen*, a Type 052D destroyer commissioned in 2017,[76]

> A ship leader admitted that the officers and soldiers have been working hard at the frontlines of the maritime struggle and the frontlines of safeguarding rights and maintaining stability all year round. "The sea is home and the port is where we are a guest" is the normal combat work of these sailors. Sometimes as soon as the ship docks, it has to prepare for the next voyage. The mission intensity is high and the transition tempo is fast, which is an unavoidable reality of the current situation.[77]

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71 OSD, 2025, pp. 89–90. The Office of the Secretary of War is designated the Department of Defense under Public Law 81-216, National Security Act Amendments of 1949.

72 OSD, 2025, pp. 89–90.

73 OSD, 2025, p. 81.

74 OSD, 2025, p. 84.

75 Lin et al., 2023; Eastern Theater WeChat Editorial Department [东部微信编辑部], "Xiangtan Ship, Taiyuan Ship, Taizhou Ship, Gather!" ["湘潭舰、太原舰、泰州舰，集合！"], WeChat [微信] social media post, April 1, 2023; Chan, 2023; "The Eastern Theater Released a Situation Map of the Multi-Directional Combat Patrol Exercise of the Warship Formation Approaching Taiwan Island" ["东部战区发布舰艇编队多方向抵近台岛战巡演练态势图"], People's Republic of China Ministry of National Defense, May 23, 2024; Dotson and Harman, 2024; Mahadzir, 2024; Liu Xin, "Update: PLA Conducts Joint Drill Surrounding Taiwan Island to Send Stern Warning to 'Taiwan Independence' Separatists," *Global Times*, October 14, 2024; Ellis, 2024; Kadidal and Rahmat, 2023.

76 Ridzwan Rahmat, "China Commissions Sixth Type 052D-Class Destroyer into East Sea Fleet," Janes, July 21, 2017.

77 Wang Yue [王悦], Shen Yang [沈扬], Ye Dengfeng [叶登峰], and Xu Wei [徐巍], "A Timely 'Brainstorming Session'" ["一场恰当其时的"头脑风暴""], *PLA Daily* [解放军报], June 13, 2024.

"Maritime struggle" likely refers to various operational missions to improve warfighting skills, and "the frontlines of safeguarding rights and maintaining stability" likely refers to presence operations that put pressure on China's neighbors in the "gray zone"—both high-end warfighting and gray zone operations are "strategic tasks" assigned to the PLAN.[78] As a 2024 article about a diesel submarine engine technician in the Eastern TCN similarly notes, the increasing normalization of long-distance combat readiness patrols has led to submarine activities becoming more frequent, and the technician "and his comrades increasingly feel that they 'cannot wait, cannot be slow, and cannot be weak'" in conducting training and combat readiness efforts.[79]

According to the information we reviewed, one unique characteristic of the **Southern TCN**'s maintenance practices is the necessity of servicing equipment stationed on China's artificial islands in the South China Sea, some of which are more than 1,000 km from mainland China. One article on a Spratlys-based naval unit notes that NCOs were developing training content and consulting technicians remotely, given that "it is inconvenient for the manufacturer's technicians [厂家技术人员] to go to the reef."[80] The article also praises NCOs who had extended their service commitments, indicating that morale or other factors might be an issue in personnel retention given the austere location. Not mentioned but likely another consideration for these far-flung units is the additional maintenance burden that high levels of salt and humidity—and severe weather more broadly—might pose.[81]

Among the three TCNs, we gleaned the fewest insights among the three TCNs about unique characteristics of **Northern TCN** activities as they pertain to maintenance. There is some evidence that ship maintainers in this fleet have opportunities to engage factory counterparts while aboard the ship [随船参加厂修培训], which could make sense given the presence of the Dalian Shipbuilding Industry Company in this theater.[82] But other fleets also have those opportunities. We also saw some evidence of emphasizing thrift or self-reliance, such as support personnel making small repairs around the barracks, both to save costs and to build skills.[83]

As one additional point of comparison, according to a list of ships participating in Gulf of Aden patrols from 2008 to early 2024, ships from all three TCNs have participated, but, of the destroyers that have conducted multiple Gulf of Aden patrols, three are from the Northern TCN, two are from the Southern TCN, and none are from the Eastern TCN.[84] Further research could examine potential variation in greater detail to determine the extent to which regional navies incorporate unique considerations into their maintenance management and the implications this could have for operations.

[78] On China's use of rights protection and stability maintenance to describe its gray zone operations, see Bonny Lin, Cristina L. Garafola, Bruce McClintock, Jonah Blank, Jeffrey W. Hornung, Karen Schwindt, Jennifer D. P. Moroney, Paul Orner, Dennis Borrman, Sarah W. Denton, and Jason Chambers, *Competition in the Gray Zone: Countering China's Coercion Against U.S. Allies and Partners in the Indo-Pacific*, RAND Corporation, RR-A594-1, 2022. On the PLAN's "strategic tasks," see Academy of Military Sciences Strategy Research Department, 2013, pp. 209–212.

[79] Ma Jialong [马嘉隆], "Soaring Through the Vast Sky, Gazing Toward the Future" ["长空振翅，眺望未来"], *PLA Daily* [解放军报], January 1, 2024.

[80] Wu Qiang [吴强], "Accelerating Along the Path of Military Service Extension" ["在军旅延长线上加速奔跑"], *PLA Daily* [解放军报], March 2, 2024.

[81] J. Michael Dahm, *Hardened Infrastructure, Counter-Reconnaissance, and Battlespace Environment Management*, Johns Hopkins Applied Physics Laboratory, 2020, p. 13.

[82] Zhou, Cheng, and Sui, 2023.

[83] Cui, Li, and Wu, 2021.

[84] From Northern TCN, these are the *Qingdao*, *Harbin*, and *Urumqi*; from Southern, the *Wuhan* and *Haikou*. In contrast, frigates from all three TCNs have participated multiple times. List of vessels from Blasko, 2024, pp. 4–7.

Partnerships Between Maintainers, Educational Institutions, and Manufacturers Demonstrate Positive Feedback Mechanisms

The PLAN is focusing on partnering with both military and civilian academic institutions. Collaboration with the civilian education system has sought to improve the quality of technical education. At the officer level, there have been investments in continuing education at the operational units, including formalized agreements with local civilian universities to provide night classes for technical specialties (e.g., Jiangsu University of Science and Technology). At the NCO level, since 2014, PLAN academic institutions have joined with civilian academic institutions to collaboratively develop NCOs in "complex specialty technical billets" (e.g., aircraft manufacturing technology and processes).[85] The intent is to create a "potentially rich source of technically proficient NCOs and one that could be readily expanded in size."[86]

Articles mention the cooperation between naval units and local universities and colleges to obtain professional qualification certificates. One article highlights a *dadui*/squadron in the Eastern TCN that partners with a local college to gain an intermediate certificate for fitters.[87] This appears similar to USN initiatives that use community colleges and vocational institutes to promote journeyman-level training. Given that the USN's lack of skilled workers prompted these sorts of initiatives, it is possible that the PLAN is facing similar challenges. Additionally, the PLAN is conducting hands-on training for students, particularly via partnerships with vocational colleges. One vocational college sent instructors to naval bases so that it could align the teaching plan with the "needs of the troops."[88] Adjustments have included adding relevant courses on maintaining gear boxes and transmission shafts, after a PLAN training base representative observed that that these parts were prone to failure but that ships lacked sufficient organic maintenance personnel with the appropriate expertise to repair them.[89]

Information-sharing between military universities and frontline forces suggests a commitment both to scientific examination of effective repair and maintenance packages and to effective training. One article about the PLAN University of Engineering notes that the university has been working on efforts to meet

> the urgent needs of combat capability generation, and has established a regular joint innovation mechanism with frontline troops, so that the latest information from the troops can be promptly transmitted to educational institutions, and the educational institutions' innovative achievements can be promptly applied to the troops, ensuring a precise alignment between technological innovation and the demands of combat readiness.[90]

Amid the rise of senior NCOs as holders of knowledge, they are being leveraged as educators to disseminate knowledge into universities about maintenance activities conducted on the front lines. In one example from the Eastern TCN, a second-class sergeant major from the PLAN's first domestically produced AIP submarine began teaching a course at the Naval Submarine Academy, later becoming a part-time professor to

[85] Kenneth Allen and Morgan Clemens, *The Recruitment, Education, and Training of PLA Navy Personnel*, China Maritime Studies Institute, 2014, p. 23.

[86] Allen and Clemens, 2014, p. 21.

[87] Liu Zhilei [刘志磊] and Xu Wei [徐巍], "Join Forces to Elevate Ships' Autonomous Support Capabilities" ["合力提升舰艇自主保障能力"], *PLA Daily* [解放军报], June 6, 2023.

[88] Wu and Liu, 2024.

[89] Wu and Liu, 2024. These examples were at Shandong Transport Vocational College [山东交通职业学院], which the article notes sent more than 300 NCOs to the PLAN in 2024.

[90] Xiong, 2021.

continue sharing insights on AIP systems with academy students.[91] These transitions can take time. When a vehicle maintainer became an instructor for a training *dadui*/squadron at the Naval Logistics Academy, the senior NCO expressed concerns about transitioning into his new role: "If I operate [the equipment] myself, I guarantee that I can do it flawlessly in one go, but explaining all these details is like 'cooking dumplings in a teapot'—I have the goods but can't get them out!"[92]

Some frontline forces also leverage partnerships with manufacturers to improve organic maintenance support. The senior engineer in one naval *dadui*/squadron, likely a destroyer or frigate flotilla, led the development of the "maintenance treasure box" mentioned earlier in this chapter, which contains common tools, spare parts, and a maintenance manual for common shipboard repairs.[93] The article notes that the engineer "repeatedly discussed and refined the plans" for the treasure box "with the military factory [军工厂]" over multiple months.[94] Although this anecdote demonstrates unit-directed innovation to solve problems, the level of close coordination required to develop a fairly straightforward maintenance kit could also indicate some level of dependency on factory technicians for the most-relevant expertise.

Organic maintainers are sometimes sent to factories to learn from their experts, indicating that one pathway to mitigate knowledge gaps is via mentoring—and large amounts of self-study, a theme that is often highlighted in the articles.[95] One master sergeant first class with no more than a junior high school education (i.e., completed ninth grade) is described as overcoming a gap in knowledge by purchasing professional books to study on his own and seeking advice from others. Specifically, "during the factory maintenance period [厂修期间], he insisted on staying on board the ship throughout the entire process, learning equipment maintenance skills from the factory's master workers [工厂师傅们], borrowing their notes, and excerpting materials to continuously enrich himself."[96] Such anecdotes as these demonstrate some tolerance for low-level mistakes and affording sailors the opportunity to learn from them. Similar to the factory personnel consulting on the treasure box, these factory masters also appear to have higher levels of technical skill and knowledge than do personnel in average naval units.

Personnel in the fleet are also identified as providing feedback to manufacturers to improve the functionality of equipment. In one example of a patrol boat operating in extreme weather in the South China Sea, the crew tested the "performance limits of the equipment" and proposed equipment modifications to the manufacturers [厂家] when issues arose with performance.[97]

These instances indicate that feedback mechanisms exist between the fleet, manufacturers, and academic institutions to help inform system design and maintenance practices.

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[91] The NCO began teaching in 2013. Fan, Liu, and Dai, 2018.

[92] Zhang Yanqing [张砚青], "From the Repair Shop to the Teaching Podium" ["从修理车间到三尺讲台"], *PLA Daily* [解放军报], April 26, 2024.

[93] The article describes a *dadui*/squadron dispatching a maintenance team [分队] to the ship to perform an emergency repair using the maintenance treasure box.

[94] Ye and Fa, 2024.

[95] Some articles describe a selection process for personnel sent from units to factories. See Zhang Yi, 2024.

[96] Gao, 2024. Per an older PLAN reference volume, a "factory repair" [工厂修理] can be "carried out on ship equipment by factories both within and outside the military" (Navy Ship Equipment Maintenance Theory Research Editorial Office, 1991, p. 6).

[97] Xu and Gao, 2024.

Morale Issues Might Have an Impact on Retention

Reporting indicates a lack of professional pride in the maintenance profession within the PLAN, similar to conditions in the PLAA and PLAAF.[98] This is especially pronounced for naval units that are shore-based rather than ship-based. Sailors in technical support *dadui*/squadrons have expressed disappointment in their assignments; as one articles describes, "becoming a support soldier meant there was no way he could board the ship, and Zou Zhiyong's heart instantly sank."[99] Similarly, recruits at naval aviation units complain that "as a naval soldier, they rarely see the sea; as a naval soldier, they have no chance to go on warships; as a naval soldier, they spend all day accompanied by mountains."[100] Some sailors are transferred from the ship to shore duty, and this transition can be difficult. Overall, sailors' inability to do the job they were hoping to do within the PLAN negatively affects morale and retention. The reporting focuses on those exceptional, or "hero," maintainers who serve despite these disappointments and perform admirably. This emphasis might be an effort to make the profession seem more noble to junior soldiers.

However, these hero maintainers seem to endure significant hardship to get the job done. The chief engineer at a Southern TCN aviation maintenance and repair facility [航保修理厂] is described as returning quickly from a medical procedure still "in diapers" to attend a naval review meeting, working diligently for five days straight.[101] Although these maintainers are examples of exemplary organic technical proficiency, they might not be the common standard within the PLAN, which could explain why their accomplishments are highlighted. Relatedly, these hero maintainers are often described as the sole sailor capable of doing specific tasks while also often taking on tasks beyond what is required of them. In one article, a maintainer is described performing low-level chores, such as painting, and he complains that new soldiers "don't understand the profession"—he worries that there will be a "'gap' in the talent pool" when he retires.[102] It appears there is some concern about the skills and proficiency of the next generation of maintainers in the PLAN, but it is difficult to assess how much of this is general complaint or a widespread problem.

Conclusion

The PLAN is investing in improving the skills of its maintenance personnel, though it perhaps has not realized improvements to organic maintainer proficiency to the degree it would like. Chapter 4 draws on insights from this chapter and the earlier shipbuilding and comparative analysis to present findings and implications for PLAN maintenance activities and management.

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98 Fleming et al., 2023.

99 Shang Jinxin [尚金鑫], Yang Chen [杨晨], and Deng Boyu [邓博宇], "The Sea Within the Hearts of the Sailors Guarding the Cave" ["守库水兵心中的海"], *PLA Daily* [解放军报], September 30, 2024.

100 Fan Enda [范恩达], Xie Fei [谢菲], and Fu Jinquan [傅金泉], "Navigator: Though I Have No Wings, My Heart Soars" ["导航兵：虽无翅膀，我心飞翔"], *PLA Daily* [解放军报], November 8, 2024.

101 Li Wei [李维] and Deng Jiajin [邓家金], "Anchor Your Original Intentions Amidst the Billowing Waves" ["把初心锚定在波涛之上"], *PLA Daily* [解放军报], April 10, 2020.

102 Su Xuesong [宿学嵩] and Wang Yuanbing [王垣镔], "'Those Who Serve in the Military with Dedication, Passion, and Love'" ["用心用情用爱当兵的人""], *PLA Daily* [解放军报], May 7, 2024.

CHAPTER 4

Findings and Implications

This report has examined the effect of the PLAN's rapid modernization on maintenance activities and management. We identified stresses and strains—some of which are self-identified in PLA and PLAN sources—brought about by this massive force structure increase and rapid modernization; these stresses and strains are creating a larger burden for the logistics support and maintenance of PLAN assets. Key factors are more-complex operations farther from China's shores, higher operational tempos, and platforms and weapon systems that are more technologically sophisticated.

Given that the PLAN is assessing the implications of these changes and how other navies have mitigated or grappled with them, the lessons the PLAN is learning could lead it to take a hybrid approach to maintenance going forward. It is likely that the PLAN will continue trying to improve its maintenance organization, processes, and culture to repair and sustain its most-capable assets (such as by overhauling its Type 052D Luyang IIIs beginning in the early 2030s) while also replacing some older classes of ships.

Findings

- Between the 1980s and the mid 2010s, the PLAN moved from a "build a little, test a little" approach of iterative design and production of small ship classes to series production of larger ship classes, such as the Type 052D Luyang III destroyer and the Type 055 Renhai cruiser (though some modifications have occurred to the Luyang III design). Series production could theoretically help the PLAN streamline maintenance requirements, processes, and associated training. However, the increased production and fielding of large surface combatants has stressed the maintenance workforce and associated organizations, such as VTCs.
- Related dynamics influence the PLAN's maintenance bow wave, which likely has yet to crest. If overhaul rates of Type 052Ds and Type 055s remain similar to those for Type 052Cs, then overhauls of China's most-capable surface combatants will begin in 2031 and last through at least 2042.
- The interplay of maintenance demand and supply linked to fleet force modernization suggests certain archetypes based on 20th- and 21st-century U.S., Japanese, and Soviet naval forces. Examples are the USN's repair-focused approach and the "boom then bust" approach of the Soviet surface fleet from the 1960s to 1980s, which fielded many new ships but was unable to maintain them. In comparison, the PLAN is more likely to take a hybrid approach as it continues to replace old force structure. The PLAN is investing in the skill of its maintenance forces and appears to be pursuing a repair-focused investment strategy. Table 4.1 summarizes this comparison.
- The PLAN has focused on improving its maintenance self-sufficiency, meaning the ability to repair and maintain equipment and systems independently of direct reach-back or broader support from external organizations. However, grassroots (those at or below the regiment level) organic maintenance forces still appear to undertake surface-level maintenance practices and depend on support from higher maintenance echelons, such as enterprise technicians, to conduct more-sophisticated repairs. This tendency could be further exacerbated if the PLAN continues to grow its force structure.

TABLE 4.1
Assessing the PLAN Against Naval Shipbuilding and Maintenance Archetypes

Navy (Period)	Arc of Force Modernization	Repair Investment Strategy	Maintenance Force Characteristics
USN surface fleet (1990–present)	Shrinking—affected by deferred maintenance and service life extensions	Repair	Skilled, but issues with manning levels
Soviet Navy surface fleet (1960–1980)	Boom then bust—emphasis on simple, cost-effective platforms	Build new	Conscript-based; poor maintenance culture
Japan MSDF submarines (2000–2020)	Steady—keeping the industrial base strong, but at higher cost	Replace	Skilled, with emphasis on a stable workforce
PLAN (2000s–present)	Rapid build—transitioning to increasingly sophisticated platforms	Transitioning to repair	Striving for self-sufficiency

- There seems to be a tension within the PLAN between promoting innovative problem-solving to improve maintenance and emphasizing dogmatic adherence to standards. Although efforts to develop a "fault-tolerant" system suggest that the PLAN seeks to destigmatize mistakes and improve knowledge-sharing, some elements of the maintenance ecosystem still call on personnel to strictly adhere to laws and regulations.
- We found some evidence that theater-specific demands might shape each TCN's maintenance management and activities, such as high operational tempos in the Eastern TCN, a need to maintain equipment on artificial islands in the South China Sea for the Southern TCN, and an emphasis on thrift and self-reliance in the Northern TCN.
- As we found in our previous work on PLAA and PLAAF maintenance forces,[1] morale issues could affect retention, particularly for naval units that are shore-based rather than ship-based. Even experienced maintainers lauded in PLA sources appear to work through exhaustion and illness to ensure that naval maintenance activities are completed.

Table 4.1 revisits our Chapter 2 comparison of naval shipbuilding and maintenance archetypes to make an assessment of the PLAN's approach.

Although the PLAN has rapidly added force structure (similar to the Soviet fleet example), it has increased the technological sophistication of each ship class while opting to conduct overhauls of relatively advanced classes, such as the Type 052C Luyang II DDGs, rather than simply replace them at mid-life with Type 052D Luyang III DDGs. The PLAN is also investing in improving the skill levels of its maintenance forces by orienting maintenance practices toward greater self-sufficiency; improving mentorship of junior organic maintainers; and deepening partnerships and knowledge sharing among maintainers, educational institutions, and manufacturers.

Implications

Of the themes we explored in this report, many are not unique to the PLAN. Issues that the USN and other navies are similarly grappling with can be grouped into the following four areas:

- **Workforce issues are trend of the times.** Although the available information about the proficiency of PLAN organic maintenance units suggests a level of dependency that the PLA is attempting to mitigate,

[1] Fleming et al., 2023.

PLAN, USN and other navies' ships are all becoming more sophisticated—meaning the bar for maintainer proficiency continues to be raised.

- **No free lunch.** Despite its impressive shipbuilding program, the PLAN does not appear to have found a way around the dynamic (also faced by other modern navies) wherein the increasing cost and sophistication of technologies being designed into new ship classes and added to existing ones have led to growing cost per ship and will thereby result in a construction slowdown—unless the PLAN or the broader PLA opts for an extreme trade-off, such as making severe cuts to procurement for other PLAN branches or other PLA services. More specifically, the PLAN seems to be pursuing a model of repair over one of replacement for large surface combatants while seeking to leverage its large and relatively new infrastructure and new technologies to advance its maintenance capabilities.
- **Concerns about cost.** The PLAN, similar to other navies, is seeking to balance costs and cost growth with desired improvements to combat effectiveness. PLA authors are studying other countries' efforts to make these trade-offs, such as service life extensions for naval ships.
- **Advanced components affecting the nature of repair.** Modern navies have different approaches to balancing repair and replacement of components and factor various elements—such as cost, supply chains, and maintainer proficiency—into their decisionmaking. The PLAN appears to be grappling with these trade-offs. To gain efficiency, navies often pursue replacement over repair (particularly for advanced components), and there is evidence that the PLAN at times adopts this approach.[2] But there is also an emphasis on repair over replacement in some reporting. Insufficient organic maintenance proficiency could be a factor constraining the PLAN from transitioning from repair to replacement. Cost might also be a factor, given that decisions about repairing advanced components versus replacing them can hinge on affordability. It is also possible that supply chain issues are driving some of the focus on repair over replacement, particularly for maintainers in remote locations. Other navies, such as the USN, face similar trade-offs but have invested in the tools and infrastructure to support these efforts. It is not clear whether the PLAN has invested similarly.

We have identified four areas that we assess as *known* or *unknown* PLAN-specific issues or concerns, meaning they are problems that the PLAN has either (1) identified but not yet addressed or (2) not yet identified (according to the sources we reviewed):

- **The sheer volume of new ships complicates maintenance pipelines and cycles (known problem).** PLA and PLAN sources recognize that downstream maintenance and sustainment activities cannot keep up with an indefinite expansion in fielded surface combatants, often described as "dumplings" arriving one after the other. The eightfold increase in one theater's VTC pipeline from 2002 to 2007, for example, reflects an effort to increase the throughput of support organizations to ensure ship deployability. Even though older ship classes are being retired, the net gain to fielded combatants has required PLAN support forces to increase maintainer efficiency—and likely training and other staff and infrastructure as well. These increased costs stack on top of the added expense of more-advanced components on new classes of surface vessels.
- **PLAN assets seem to undergo significant changes that require maintainers to reskill or develop new resource materials from scratch (a potentially unknown problem).** We found multiple references to specific ships or vehicles adding new or upgraded versions of existing systems, which increased the complexity of maintenance and repair tasks. Even the most-capable hero maintainers discussed needing to remap pipeline pathways after each significant repair on their ship, for example. This might tie

2 We saw this referenced by a helicopter maintainer at a university. Zhang and Luo, 2024.

back to one article's criticism of a "monopolistic repair model" in which each system developer fulfills the specified requirements, but the authors noted a lack of integration across stakeholders that might impose additional burdens on organic maintenance personnel.[3] Documentation failures of this type can sometimes occur in the USN and in other navies, but this does appear to be an issue exacerbating maintainers' workloads across the PLAN.

- **A focus on achieving self-sufficiency could be counterproductive (potentially not recognized as a problem).** As systems become increasingly technical and complex, nonorganic support from enterprise technicians might be in even higher demand. Organic maintenance expertise not keeping pace with technically complex new classes of equipment is a common issue in many military organizations, including the USN. However, increasing maintenance self-sufficiency appears to be a long-standing and idealized goal for the PLAN. Pursuing self-sufficiency to an extreme could reduce overall efficiency, particularly since the PLAN already identifies a lack of organic maintenance capacity as a shortfall. Whether the PLAN decides to strike some kind of balance between self-sufficiency and efficiency remains to be seen.
- **Emphasis on repair over replacement for small components could waste effort instead of maximizing efficiency (potentially not recognized as a problem).** Allocating personnel time to repair small components, such as light bulb capacitors, might leave them less time to engage in more-sophisticated training opportunities, mentoring junior personnel, or other activities. A potential exception might be if these units are facing supply issues, which would raise broader questions about PLAN investment in a more-capable maintainer force; we did not see specific evidence of this one way or the other. These stories could also be highlighted for propaganda effect more than operational utility; however, if this behavior were observed across units and TCNs, it might reduce the broader efficiency of PLAN maintenance activities.

Future Outlook and Conclusion

Our analysis suggests some competing dynamics for how military logistics might evolve and the implications for naval maintenance. Although such advances as improved data fusion, agentic tools, additive manufacturing, and robotics could significantly reshape the nature of organic maintenance support, wartime conditions could still result in degraded information environments or other system shortfalls that would complicate the employment of next-generation tools. As one *PLA Daily* article puts it, despite the potential impact of these innovations on logistics, "it is still necessary to leverage human initiative and continuously innovate support methods to meet combat needs."[4] One possible tension stems from whether the adoption of artificial intelligence–related tools would exacerbate or lessen dependency on reach-back support and whether this might create additional dependencies that organic maintainers would need to work around if the tool goes offline. Another area to consider is the extent to which these tools or other process improvements could help mitigate both the PLAN's expected bow wave in maintenance (once the next major tranche of overhauls begins in the early 2030s) and the associated stresses and strains on such organizations as VTCs, which ensure ship deployability.

Additional research could further examine (1) these implications; (2) related topics, such as maintenance in other naval branches (especially submarine and aircraft carriers and carrier aviation); and (3) other subfunctions of logistics that intersect with maintenance, such as supply.

3 Zhang and Zhu, 2023, pp. 45–46.

4 Song Youjian [宋永健], "Speaking of Logistics Support via 'Lychees from Chang'an'" ["长安的荔枝" 话后勤保障"], *PLA Daily* [解放军报], November 6, 2025.

APPENDIX

Launch and Commissioning Dates of PLAN Large Surface Combatants, 2000–2025

Table A.1 lists the known launch and commissioning dates for the PLAN's largest surface combatants—aircraft carriers, cruisers, and destroyers—based on information from Janes updated as of January 26, 2026. Ships are grouped by type and listed in order of launch date. All information comes from Janes unless otherwise noted.

TABLE A.1

Launch and Commissioning Dates of PLAN Large Surface Combatants Since 2000

Type	Pennant Number	Ship Name	Builder	Class	Launch Date	Commissioning Date	Displacement (Full Load, Tonnes)
CVM	16	*Liaoning*	Nikolayev South/Dalian Shipyard	Type 001	December 4, 1988	September 25, 2012	58,500
CVM	17	*Shandong*	Dalian Shipyard	Type 002	April 26, 2017	December 17, 2019	70,000
CVM	18	*Fujian*	Jiangnan Shipyard, Changxingdao	Type 003	June 17, 2022	November 5, 2025	80,000
CG	101	*Nanchang*	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	June 28, 2017	January 12, 2020	13,000
CG	102	*Lasa*	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	April 28, 2018	March 2, 2021	13,000
CG	103	*Anshan*	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	September 12, 2019	November 11, 2021	13,000
CG	104	*Wuxi*	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	May 9, 2020	March 10, 2022[a]	13,000
CG	105	*Dalian*	Dalian Shipbuilding Industry Company	Renhai/ Type 055	July 3, 2018	April 23, 2021	13,000
CG	106	*Yan'an*	Dalian Shipbuilding Industry Company	Renhai/ Type 055	July 3, 2018	February 22, 2022[a]	13,000
CG	107	*Zunyi*	Dalian Shipbuilding Industry Company	Renhai/ Type 055	December 26, 2019	November 2022	13,000
CG	108	*Xianyang*	Dalian Shipbuilding Industry Company	Renhai/ Type 055	August 30, 2020	December 2022[b]	13,000
CG	109	*Dongguan*	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	December 28, 2023	2025	13,000
CG	110	*Anqing*	Dagushan Shipyard, Dalian	Renhai/ Type 055	March 26, 2024	Expected 2026	13,000
CG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	March 2025	Unknown	13,000
CG	Unknown	Unknown	Dagushan Shipyard, Dalian	Renhai/ Type 055	April 2025	Unknown	13,000
CG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Renhai/ Type 055	September 2025	Unknown	13,000

Table A.1—Continued

Type	Pennant Number	Ship Name	Builder	Class	Launch Date	Commissioning Date	Displacement (Full Load, Tonnes)
CG	Unknown	Unknown	Dagushan Shipyard, Dalian	Renhai/ Type 055	October 2025	Unknown	13,000
DDG	168	*Guangzhou*	Jiangnan Shipyard, Shanghai	Luyang I/ Type 052B	May 25, 2002	July 15, 2004	6,500
DDG	169	*Wuhan*	Jiangnan Shipyard, Shanghai	Luyang I/ Type 052B	September 9, 2002	December 2004	6,500
DDG	170	*Lanzhou*	Jiangnan Shipyard, Shanghai	Luyang II/ Type 052C	April 29, 2003	October 18, 2005	7,000
DDG	171	*Haikou*	Jiangnan Shipyard, Shanghai	Luyang II/ Type 052C	October 30, 2003	December 26, 2005	7,000
DDG	115	*Shenyang*	Dalian Shipyard	Luzhou/ Type 051C	December 28, 2004	November 28, 2006	7,000
DDG	116	*Shijiazhuang*	Dalian Shipyard	Luzhou/ Type 051C	July 26, 2005	January 17, 2007	7,000
DDG	150	*Changchun*	Jiangnan Shipyard, Changxingdao	Luyang II/ Type 052C	November 28, 2010[C]	January 31, 2013	7,000
DDG	151	*Zhengzhou*	Jiangnan Shipyard, Changxingdao	Luyang II/ Type 052C	July 20, 2011	December 26, 2013	7,000
DDG	152	*Jinan*	Jiangnan Shipyard, Changxingdao	Luyang II/ Type 052C	October 16, 2011[C]	December 22, 2014	7,000
DDG	153	*Xi'an*	Jiangnan Shipyard, Changxingdao	Luyang II/ Type 052C	May 28, 2012[C]	February 9, 2015	7,000
DDG	172	*Kunming*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 28, 2012	March 21, 2014	7,500
DDG	173	*Changsha*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 18, 2012	August 12, 2015	7,500
DDG	174	*Hefei*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	July 1, 2013	December 12, 2015	7,500
DDG	175	*Yinchuan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	March 28, 2014	July 12, 2016	7,500

Table A.1—Continued

Type	Pennant Number	Ship Name	Builder	Class	Launch Date	Commissioning Date	Displacement (Full Load, Tonnes)
DDG	117	*Xining*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 26, 2014	January 22, 2017	7,500
DDG	154	*Xiamen*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 30, 2014	June 10, 2017	7,500
DDG	118	*Urumqi*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	July 7, 2015	January 2018	7,500
DDG	119	*Guiyang*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	November 28, 2015	February 22, 2019	7,500
DDG	155	*Nanjing*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 28, 2015	April 2, 2018[a]	7,500
DDG	131	*Taiyuan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	July 28, 2016	December 3, 2018	7,500
DDG	120	*Chengdu*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	August 3, 2016	November 22, 2019	7,500
DDG	161	*Huhehaote*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 26, 2016	January 12, 2019	7,500
DDG	121	*Qiqiha'er*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	June 26, 2017	August 14, 2020	7,500[d]
DDG	122	*Tangshan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	July 7, 2018	August 14, 2020	8,000[d]
DDG	156	*Zibo*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 14, 2018	January 12, 2020	8,000
DDG	132	*Suzhou*[e]	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 18, 2018	January 15, 2021[a]	8,000
DDG	162	*Nanning*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	February 23, 2019	April 12, 2021	8,000
DDG	123	*Huainan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	April 15, 2019	February 2021	8,000
DDG	124	*Kaifeng*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	May 10, 2019	April 16, 2021	8,000

Table A.1—Continued

Type	Pennant Number	Ship Name	Builder	Class	Launch Date	Commissioning Date	Displacement (Full Load, Tonnes)
DDG	165	*Zhanjiang*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	May 10, 2019	December 25, 2021	8,000
DDG	133	*Baotou*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 28, 2019	December 28, 2021	8,000
DDG	164	*Guilin*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	September 26, 2019	September 2021	8,000
DDG	134	*Shaoxing*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	December 26, 2019	February 2022	8,000
DDG	163	*Jiaozuo*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 28, 2019	January 17, 2022[a]	8,000
DDG	157	*Lishui*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	August 30, 2020	June 2022	8,000
DDG	135	*Dazhou*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	December 28, 2022[a]	2024	8,000
DDG	166	*Weinan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	2023[b]	2025	8,000
DDG	125	*Cangzhou*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	March 10, 2023	2025	8,000
DDG	126	*Heze*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	March 10, 2023	2025	8,000
DDG	158	*Suzhou*[e]	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	April 2023[b]	2025	8,000
DDG	127	*Zhumadian*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	August 2023	2025	8,000
DDG	176	*Loudi*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	January 28, 2024	2025	8,000
DDG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	March 2024	Unknown	8,000
DDG	128	*Ganzi*	Dalian Shipyard	Luyang III/ Type 052D/DL/DM	May 17, 2024	November 2025	8,000

Table A.1—Continued

Type	Pennant Number	Ship Name	Builder	Class	Launch Date	Commissioning Date	Displacement (Full Load, Tonnes)
DDG	177	*Tongchuan*	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 2024	Expected 2026	8,000
DDG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 2024	Unknown	8,000
DDG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	March 2025	Unknown	8,000
DDG	Unknown	Unknown	Dagushuan Shipyard, Dalian	Luyang III/ Type 052D/DL/DM	April 2025	Unknown	8,000
DDG	Unknown	Unknown	Jiangnan Shipyard, Changxingdao	Luyang III/ Type 052D/DL/DM	August 27, 2025	Unknown	8,000

SOURCES: Features information from Janes, “Luyang I (Type 052B) Class (DDGHM),” webpage, December 11, 2025g; Janes, “Luzhou (Type 051C) Class (DDGHM),” webpage, September 30, 2025d; Janes, “Luyang II (Type 052C) Class (DESTROYERS) (DDGHM),” webpage, July 2, 2025a; Janes, 2026a; Janes, 2026e; Janes, 2026c; Janes, 2025b; Janes, 2025f; Janes, “Pennant Numbers of Major Surface Ships in Numerical Order,” webpage, October 31, 2025e; Janes, 2025c; Ju Lang, 2024, p. 1; Seaforces.org, “Type 052D Luyang III Class Guided Missile Destroyer,” webpage, undated-a; Seaforces.org, “Type 055 Renhai Class Guided Missile Destroyer,” webpage, undated-b.

[a] In some cases, we found that launch and commissioning dates from Seaforces.org matched those from Janes but added the specific day of the month to the month and year listed by Janes. These additions are reflected in the table and apply to commissioning dates for the cruisers *Wuxi* (104) and *Yan’an* (106); the destroyers *Nanjing* (155), *Suzhou* (132), and *Jiaozuo* (163); and the destroyer *Dazhou* (135).

[b] For three other ships, Seaforces.org lists their launch or commissioning dates slightly differently than those from Janes; we were not able to account for the discrepancy. These ships are the cruiser *Xianyang* (108) and the destroyers *Weinan* (166) and *Suzhou* (158). Seaforces.org lists the commissioning date for the *Xianyang* as April 21, 2023 (not December 2022), the launch date for the *Weinan* as January 28, 2024 (not 2023); and the launch date for the *Suzhou* as August 28, 2023 (not April 2023). See Seaforce.org in the SOURCES above.

[c] For three Type 052C destroyers, a PLA source lists their launch dates slightly differently than Janes; we were not able to account for the discrepancy. These ships are the *Changchun* (pennant number 150), *Jinan* (152), and *Xi’an* destroyers (153), for which the PLA source has their launch dates as October 2010, January 2012, and July 2012, respectively. See Ju Lang, 2024, p. 1.

[d] On the differences in tonnage among Type 052D variants, see Janes, 2026a.

[e] Both DDG 132 and DDG 158 are named *Suzhou*, but after different cities in China; 苏州 for DDG 132 and likely 宿州 for DDG 158.

Abbreviations

AIP	air-independent propulsion
CCP	Chinese Communist Party
CG	guided missile cruiser
CGHM	cruiser with surface-to-surface missiles, a hangar, and surface-to-air missiles
CVM	aircraft carrier
DDG	guided missile destroyer
DDGHM	destroyer with surface-to-surface missiles, a hangar, and surface-to-air missiles
FFGHM	frigate with surface-to-surface missiles, a hangar, and surface-to-air missiles
GAO	U.S. Government Accountability Office
MSDF	Maritime Self-Defense Force (Japan)
NCO	noncommissioned officer
OSD	Office of the Secretary of Defense
PLA	People's Liberation Army
PLAA	People's Liberation Army Army
PLAAF	People's Liberation Army Air Force
PLAN	People's Liberation Army Navy
PLAN HQ	People's Liberation Army Navy Headquarters
PLANMC	People's Liberation Army Navy Marine Corps
PRC	People's Republic of China
TCN	Theater Command Navy
USN	U.S. Navy
VTC	vessel training center

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The ability for navies to procure, field, and sustain large surface combatants is central to modern power projection. The U.S. Navy's approach to ship procurement coupled with infrastructure issues and workforce shortages at U.S. shipyards have led to significant delays in both naval shipbuilding and maintenance activities. Tensions between high operational demands and the need to conduct training and maintenance to sustain ship readiness levels have resulted in significant mishaps and crashes. The ability of the People's Liberation Army Navy (PLAN) to sustain its growing inventories of more-capable systems will rely on its logistics capabilities and processes. Understanding the PLAN's approach to maintenance management is a necessary step in assessing the People's Liberation Army's ability to sustain a modernized force, and there are implications for both competition and conflict.

In this report, the authors examine how the rapid growth and modernization of the PLAN inform maintenance management and activities, approaches that other navies have taken to balance maintenance demand and supply and how the PLAN's approaches compare with them, how maintenance is controlled and organized to meet maritime-specific requirements, and the factors that shape the PLAN's ability to maintain and sustain its forces.

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